



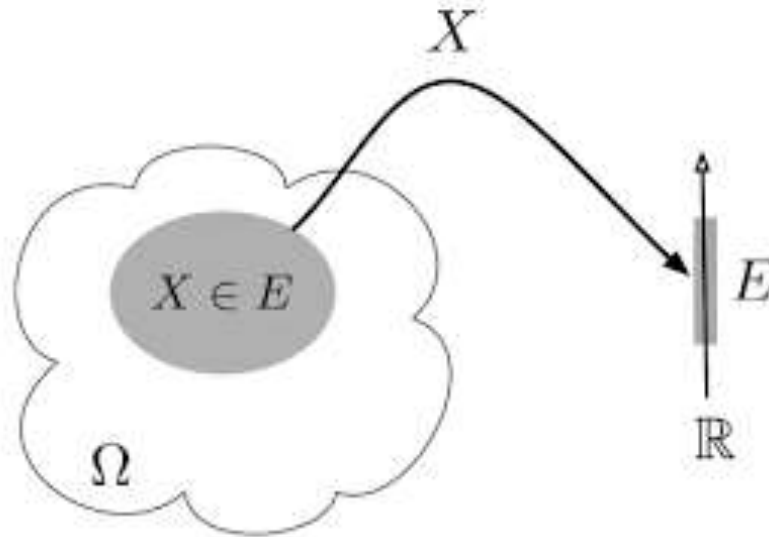
Probability Distributions



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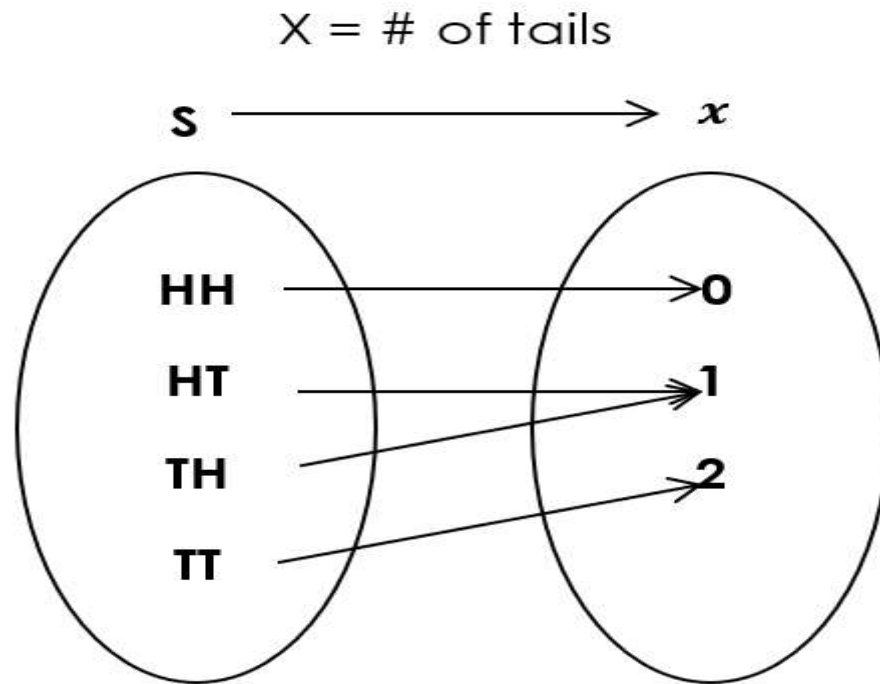
Random Variables

- A random variable is a function that assigns a numerical value to each possible outcome of a random experiment.

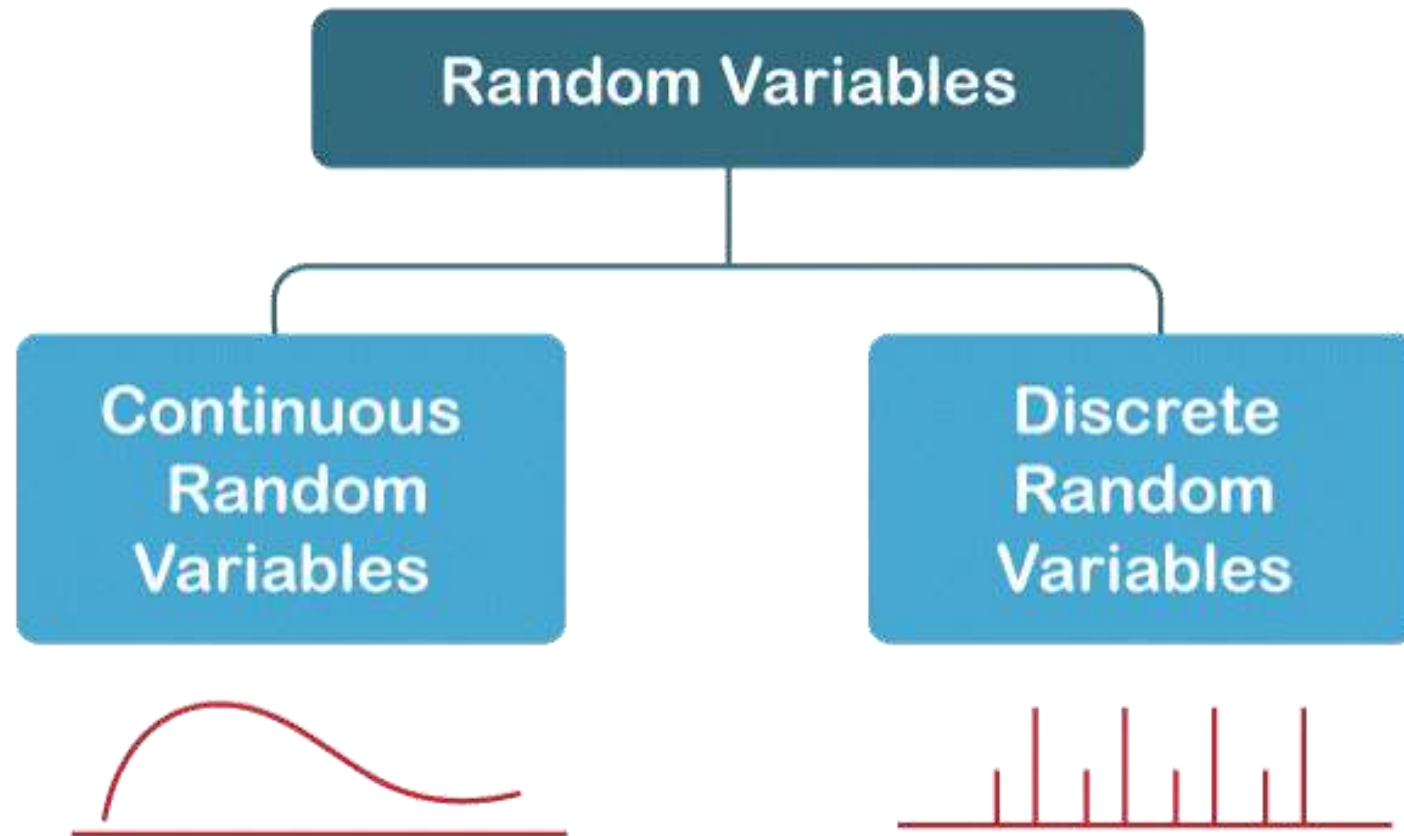


Random Variables

- Ex- X = number of tails obtained when tossing a coin once.



Types of Random Variables



Discrete Random Variables Examples

- Tossing a Coin Three Times
- Sample Space (S):
 - $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$
- Let X = number of heads obtained
- $X \in \{0, 1, 2, 3\}$

Discrete Random Variables Examples

Outcome	X
HHH	3
HHT	2
HTH	2
THH	2
HTT	1
THT	1
TTH	1
TTT	0

Discrete Random Variables Examples

- Rolling a Die Once
- Sample Space (S):
 - $S=\{1,2,3,4,5,6\}$
- Let X = number shown on the die
- $X \in \{1,2,3,4,5,6\}$

Discrete Random Variables Examples

Outcome	X
1	1
2	2
3	3
4	4
5	5
6	6

Discrete Random Variables Examples

- Inspecting 2 Light Bulbs (Defective or Not)
- Sample Space (S):
 - $S = \{DD, DN, ND, NN\}$
- Let X = number of defective bulbs
- $X \in \{0, 1, 2\}$

Discrete Random Variables Examples

Outcome	X
DD	2
DN	1
ND	1
NN	0

Continuous Random Variables Examples

- Height of a Student
- Sample Space (S):
 - $S = \{x \in \mathbb{R} \mid 140 \leq x \leq 190\}$
- Let X = height of a student (in cm)
- $X \in [140, 190]$

Continuous Random Variables Examples

- Time Taken to Complete an Exam
- Sample Space (S):
 - $S = \{x \in \mathbb{R} \mid 0 < x \leq 180\}$
- Let X = time taken to complete an exam (in minutes)
- $X \in (0, 180]$

Continuous Random Variables Examples

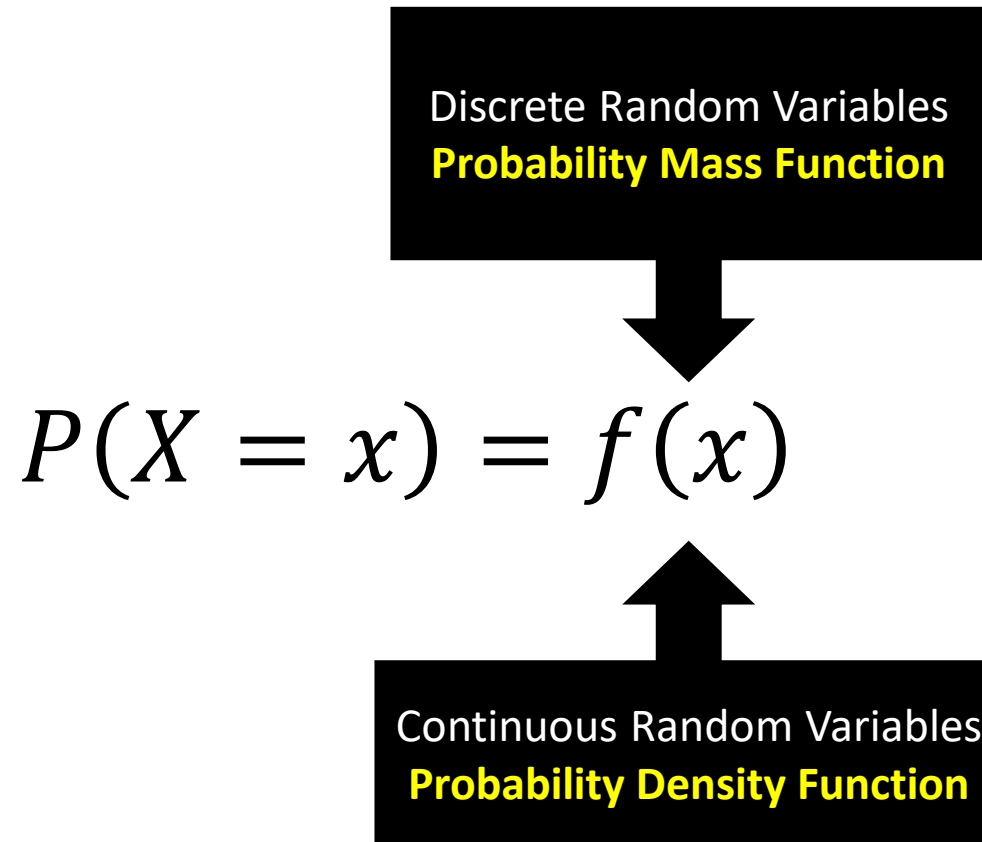
- Weight of a Fruit
- Sample Space (S):
 - $S = \{x \in \mathbb{R} | 0.2 \leq x \leq 1.5\}$
- Let X = weight of a mango (in kg)
- $X \in [0.2, 1.5]$

Probability Associated with Random Variables

- Using the values of random variables, associated probabilities can be found.
- Ex:- An experiment consists of tossing two fair coins simultaneously
 - $S = \{(H,H), (H,T), (T,H), (T,T)\}$
- X = Number of Heads Obtained (0, 1, 2)
- Ex – $P(x = 1) = \frac{2}{4} = \frac{1}{2}$
- Ex – $P(x \geq 1) = P(x = 1) + P(x = 2) = \frac{3}{4}$

Probability Associated with Random Variables

- Probability of a random variable with any value can be represented in a functional form.



Probability Distribution

- Any function has a distribution or an arrangement with its inputs and outputs.

X	f(X)
a	f(a)
b	f(b)
c	f(c)

- In a random variable, this arrangement is called as a Probability Distribution.
- With the nature of the random variable, the distribution can be a **Discrete Probability Distribution** or a **Continuous Probability Distribution**.

Discrete Probability Distributions

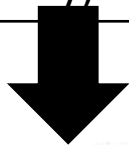
- Ex- Toss a fair coin twice
- X = number of heads obtained

X	0	1	2
$P(X=x)$ (Probability)	$1/4$	$2/4$	$1/4$

Discrete Probability Distributions

- Ex- Toss a fair coin twice
- X = number of heads obtained

X	0	1	2
$P(X=x)$ (Probability)	$1/4$	$2/4$	$1/4$
$F(X=x)$ (Cumulative Probability)	$1/4$	$3/4$	$4/4 = 1$



$$P(X \leq x) = \sum P(X = x_i)$$

Discrete Probability Distributions

- The probability of a discrete random variable lies between 0 and 1:
 - $0 \leq P(X = x) \leq 1$
- Sum of Probabilities is always equal to 1:
 - $\sum P(X = x) = 1$

Continuous Probability Distributions

$$P(a \leq x \leq b) = \int_a^b f(x)dx \geq 0$$

- All properties are common for continuous probability distributions too.

Continuous Probability Distributions

$$f_X(x) = \begin{cases} cx^2, & -1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Questions:

1. Find the value of c .
2. Find $Pr(0 < X \leq 1)$.

Expected Value

- Discrete

$$\mathbb{E}(X) = \sum_x x f_X(x) = \sum_x x \mathbb{P}(X = x).$$

- Continuous

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Variance

- Discrete or Continuous

$$\text{Var}(X) = \mathbb{E}\left((X - \mu_X)^2\right) = \mathbb{E}(X^2) - \left(\mathbb{E}(X)\right)^2.$$

Expected Value and Variance

- Find the $E(X)$ and $V(X)$

X	0	1	2	3
P_x	1/10	2/10	4/10	3/10

Expected Value and Variance

$$f_X(x) = \begin{cases} cx^2, & -1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Questions:

1. Find the value of c .
2. Find $Pr(0 < X \leq 1)$.
3. Find the expected value $E(X)$ and the variance $Var(X)$.
4. Find the cumulative distribution function (c.d.f.).

Discrete Uniform Distribution

- A random variable is defined to have a discrete uniform distribution, if the probability mass function of X is given by,

$$P_X(x) = \begin{cases} \frac{1}{N}, & \text{for } i = 1, 2, \dots, N \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = \frac{N+1}{2} \text{ and } V(X) = \frac{N^2-1}{12}$$

Discrete Uniform Distribution

- Example: Let X represent a random variable taking on the possible values of 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, and each possible value has equal probability.
- This is a discrete uniform distribution and the probability for each of the 10 possible value is

$$P(X = x_i) = f(x_i) = \frac{1}{10} = 0.10$$

Bernoulli Distribution

- Consider the toss of a coin, which comes up a head with probability p , and a tail with probability $1-p$. Define

$$X = \begin{cases} 1, & \text{if Head} \\ 0, & \text{if Tail} \end{cases}$$

- The random variable X is said to have a Bernoulli distribution and the probability mass function of X is given by,

$$P_X(x) = \begin{cases} p^x(1-p)^{1-x}, & \text{for } x=0 \text{ or } 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = p \text{ and } V(X) = p(1-p)$$

Binomial Distribution

- A coin is tossed n times. At each toss, the coin comes up a head with probability p , and a tail with probability $1-p$, independently of prior tosses.
- Let X be the number of heads in the n -toss sequence.
- The random variable X is said to have a Binomial distribution with parameters n and p and the probability mass function of X is given by,

$$P_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{(n-x)}, & \text{for } x = 0, 1, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Also, $E(X) = np$ and $V(X) = np(1-p)$

Binomial Distribution

- Example: A die is tossed 3 times. What is the probability of
 - No fives turning up?
 - 1 five?
 - 3 five?

Binomial Distribution

This is a **binomial** distribution because there are only 2 possible outcomes (we get a 5 or we don't).

Now, $n = 3$ for each part. Let X = number of fives appearing.

(a) Here, $x = 0$.

$$P(X = 0) = C_x^n p^x q^{n-x} = C_0^3 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = \frac{125}{216} = 0.5787$$

(b) Here, $x = 1$.

$$P(X = 1) = C_x^n p^x q^{n-x} = C_1^3 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = \frac{75}{216} = 0.34722$$

(c) Here, $x = 3$.

$$P(X = 3) = C_x^n p^x q^{n-x} = C_3^3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = \frac{1}{216} = 4.6296 \times 10^{-3}$$

Hypergeometric Distributions

- Assume that a population of size N which contains M defective items.
- A random sample of size n is drawn and observed that it contains X number of defective items. The random variable X is said to have a hypergeometric distribution and the probability mass function of X is given by

$$P_X(x) = \begin{cases} \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}, & \text{for } x = 0, 1, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = n \frac{M}{N} \text{ and } V(X) = \frac{nK}{N} \left(1 - \frac{K}{N} \right) \frac{N-n}{N-1}$$

Hypergeometric Distributions

- Example: Suppose we randomly select 5 cards without replacement from an ordinary deck of playing cards. What is the probability of getting exactly 2 red cards (i.e., hearts or diamonds)?
 - $N = 52$; since there are 52 cards in a deck.
 - $M = 26$; since there are 26 red cards in a deck.
 - $n = 5$; since we randomly select 5 cards from the deck.
 - $x = 2$; since 2 of the cards we select are red.

We plug these values into the hypergeometric formula

Thus, the probability of randomly selecting 2 red cards is 0.32513.

Poisson Distributions

- The number of events occurring (X) within a fixed period of time has Poisson distribution and its probability mass function is given by

$$P_X(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{for } x = 0, 1, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Also, $E(X) = \lambda$ and $V(X) = \lambda$

- Poisson distribution can be effectively used to approximate Binomial probabilities when the number of trials n is large, and the probability of success p is small.

Poisson Distributions

- Example: A life insurance salesman sells on the average 3 life insurance policies per week. Use Poisson's law to calculate the probability that in a given week he will sell
 - Some policies
 - 2 or more policies but less than 5 policies.

Poisson Distributions

Here, $\mu = 3$

(a) "Some policies" means "1 or more policies". We can work this out by finding 1 minus the "zero policies" probability:

$$P(X > 0) = 1 - P(x_0)$$

$$\text{Now } P(X) = \frac{e^{-\mu} \mu^x}{x!} \text{ so } P(x_0) = \frac{e^{-3} 3^0}{0!} = 4.9787 \times 10^{-2}$$

Therefore the probability of 1 or more policies is given by:

$$\text{Probability} = P(X \geq 0)$$

$$= 1 - P(x_0)$$

$$= 1 - 4.9787 \times 10^{-2}$$

$$= 0.95021$$

(b) The probability of selling 2 or more, but less than 5 policies is:

$$P(2 \leq X < 5)$$

$$= P(x_2) + P(x_3) + P(x_4)$$

$$= \frac{e^{-3} 3^2}{2!} + \frac{e^{-3} 3^3}{3!} + \frac{e^{-3} 3^4}{4!}$$

$$= 0.61611$$

Geometric Distribution

- Suppose that we repeatedly and independently toss a coin with probability of a head p .
- The geometric random variable is the number X of tosses needed before a head to come up for the first time. The probability mass function of X is given by,

$$P_X(x) = \begin{cases} p(1-p)^{x-1} & \text{for } x = 1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = \frac{1}{p} \quad \text{and} \quad V(X) = \frac{1-p}{p^2}$$

Geometric Distribution

- Example: A farmer tests milk quality from cows one by one. Each test has a probability of passing equal to 0.2. Tests are independent. Probability that the first success occurs on the 4th test

$$\begin{aligned}P(X = 4) &= (1 - 0.2)^3 \times 0.2 \\&= (0.8)^3 \times 0.2 \\&= 0.1024\end{aligned}$$

Negative Binomial Distribution

- Let X denote the number of trials until the r^{th} success.
- Then, the probability mass function of X is

$$P_X(x) = \begin{cases} \binom{x-1}{r-1} (1-p)^{x-r} p^r, & \text{for } x = r, r+1, r+2, \dots \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = \frac{r}{p} \text{ and } V(X) = \frac{r(1-p)}{p^2}$$

Negative Binomial Distribution

- Example: A dairy quality inspector tests milk samples one by one. Each test has a probability of being acceptable equal to 0.3. Tests are independent. Probability that the 3rd acceptable sample occurs on the 7th test

$$\begin{aligned}P(X = 7) &= \binom{6}{2} (0.3)^3 (0.7)^4 \\&= 15 \times 0.027 \times 0.2401 \\&= 0.0972 \text{ (approximately)}\end{aligned}$$

Continuous Uniform Distribution

- A random variable is defined to have a uniform distribution, if the probability density function of X is given by

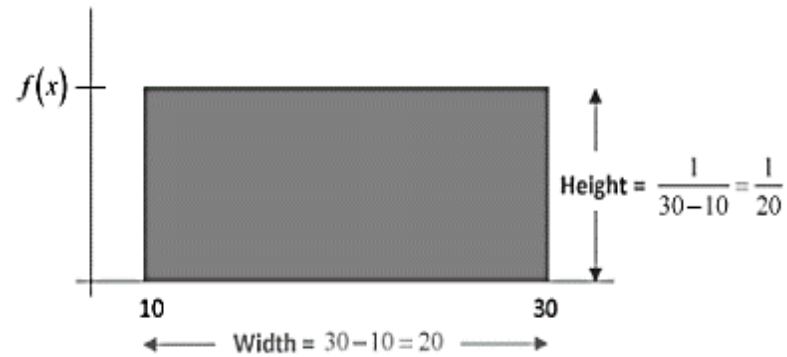
$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{for } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Also, } E(X) = \frac{a+b}{2} \text{ and } V(X) = \frac{(b-a)^2}{12}$$

Uniform Distribution

- Imagine you live in a building that has an elevator that will take you to your floor.
- From experience, once you push the button to call the elevator, it takes between ten and thirty seconds for you to arrive at your floor.
- This means the elevator arrival is uniformly distributed between 10 and 30 seconds once you hit the button.
- So, if X is a continuous uniform random variable has probability density function mean, and variance is as follows.

Uniform Distribution



Probability Density Function

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{elsewhere} \end{cases} \rightarrow f(x) = \begin{cases} \frac{1}{20} & 10 \leq x \leq 30 \\ 0 & \text{elsewhere} \end{cases}$$

Mean

$$\mu = E(X) = \int x f(x) dx \rightarrow \int_{10}^{30} x \left(\frac{1}{20} \right) dx = \int_{10}^{30} \left(\frac{1}{20} x \right) dx = \frac{1}{40} x^2 \Big|_{10}^{30} = 20$$

or

$$\mu = E(X) = \frac{b+a}{2} \rightarrow E(X) = \frac{30+10}{2} = 20$$

Variance

$$\sigma^2 = V(X) = \int (x - \mu)^2 f(x) dx \rightarrow \int_{10}^{30} (x - 20)^2 \left(\frac{1}{20} \right) dx = \int_{10}^{30} \left(\frac{1}{20} x^2 - 2x + 20 \right) dx = \frac{100}{3} \approx 33.33$$

or

$$\sigma^2 = V(X) = \frac{(b-a)^2}{12} \rightarrow V(X) = \frac{(30-10)^2}{12} = \frac{100}{3} \approx 33.33$$

Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{33.33} = 5.774$$

Normal Distribution

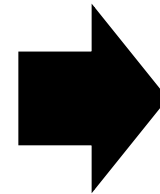
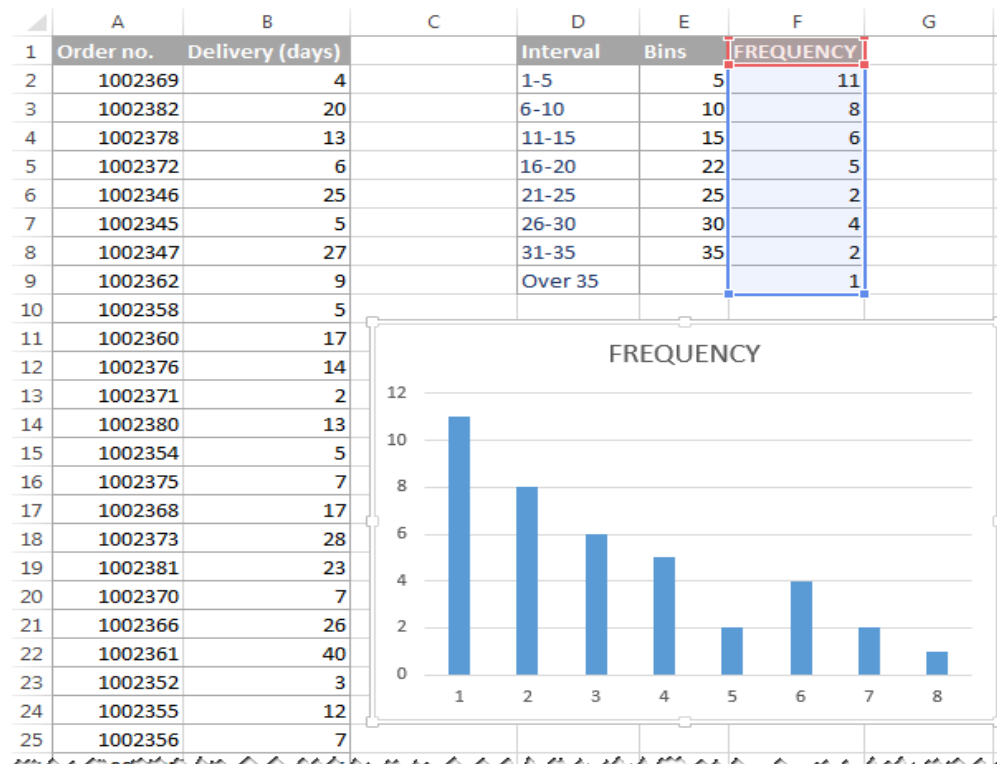
- A random variable is defined to have a normal distribution with mean μ and variance σ^2 , if the probability density function of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty \leq x \leq \infty$$

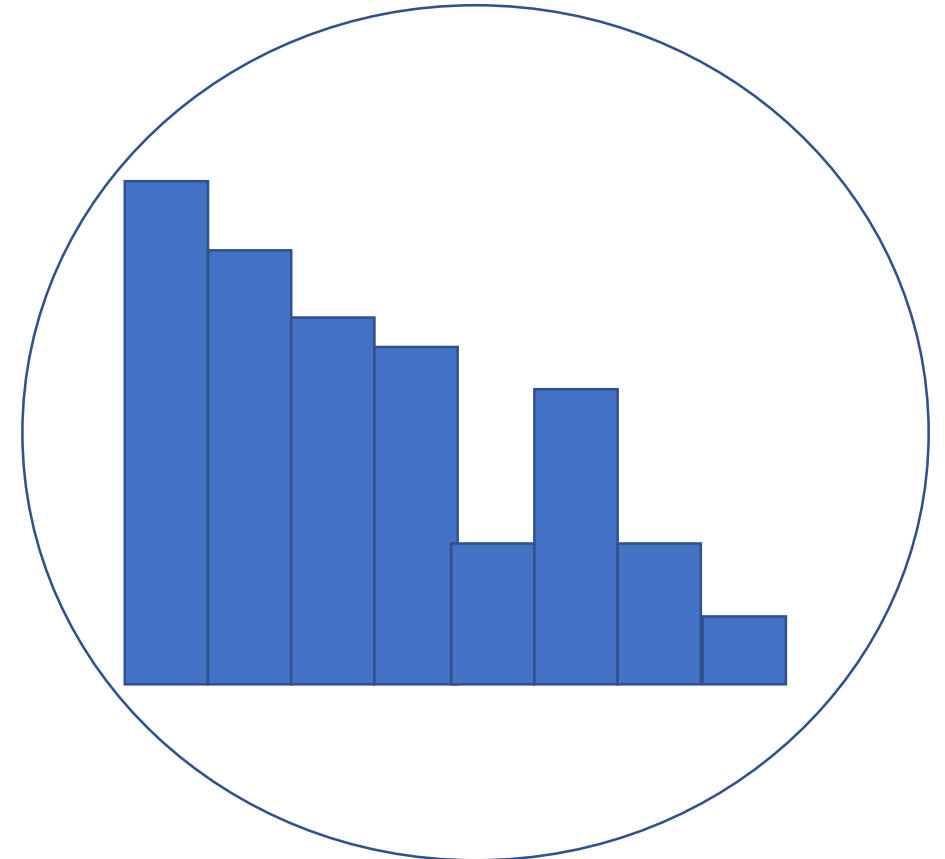
$$\text{Also, } E(X) = \mu \text{ and } V(X) = \sigma^2$$

Normal Distribution

The distribution of a statistical data set (or a population) is a listing or function showing all the possible values (or intervals) of the data and how often they occur. Consider the following example



Distribution

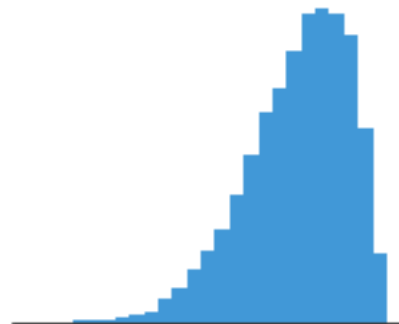


Normal Distribution

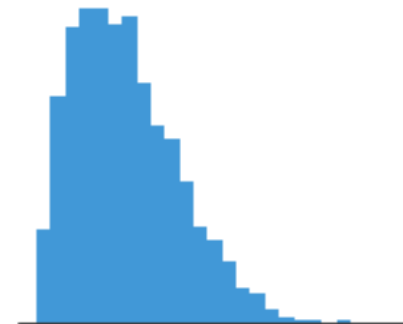
There are several shapes of these distributions.



symmetric, unimodal



skew left



skew right



uniform

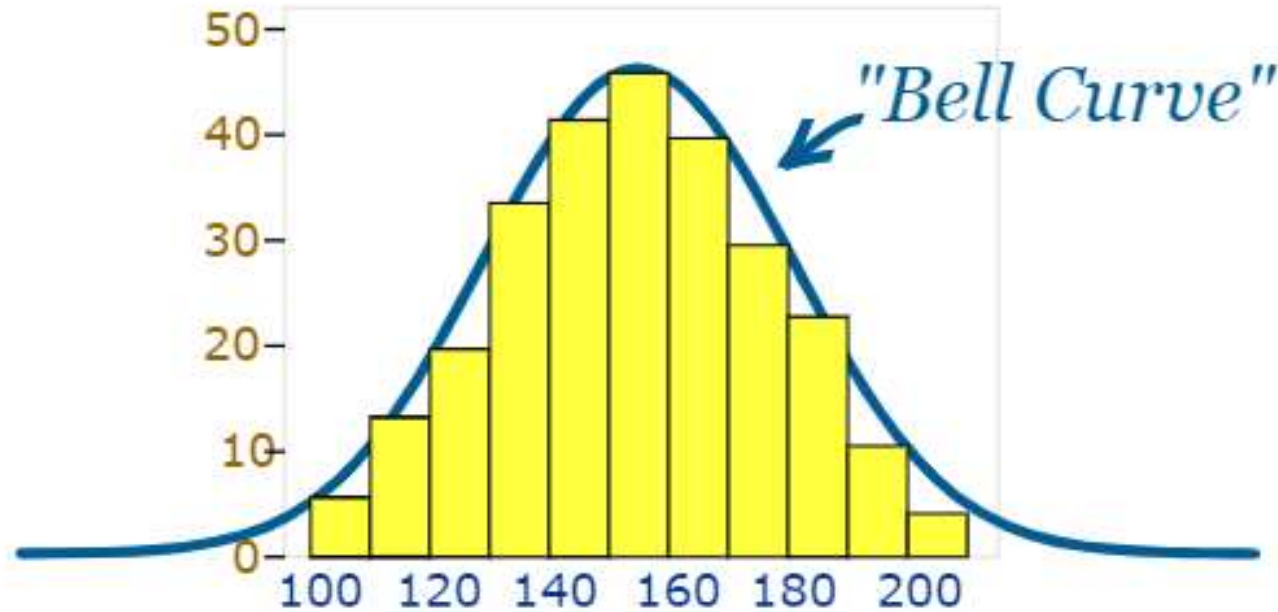


bimodal



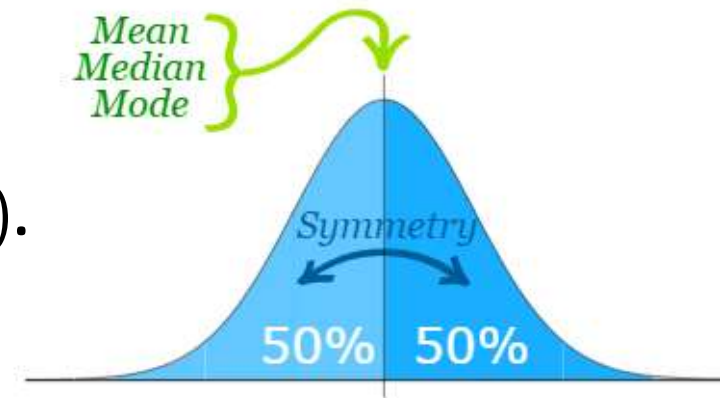
multimodal

Normal Distribution

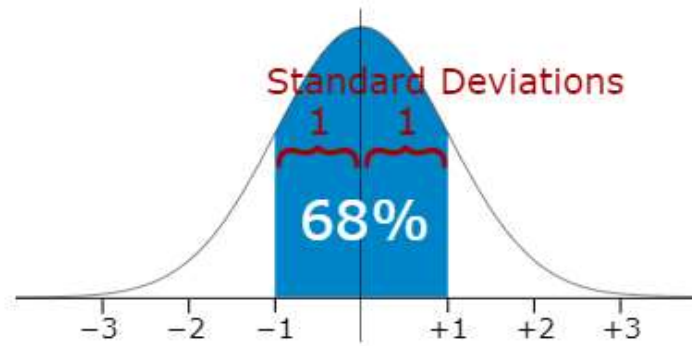


Important parameters are the **mean**(μ) and the **variance** (σ^2).
Standard deviation (σ) is the square root of variance.

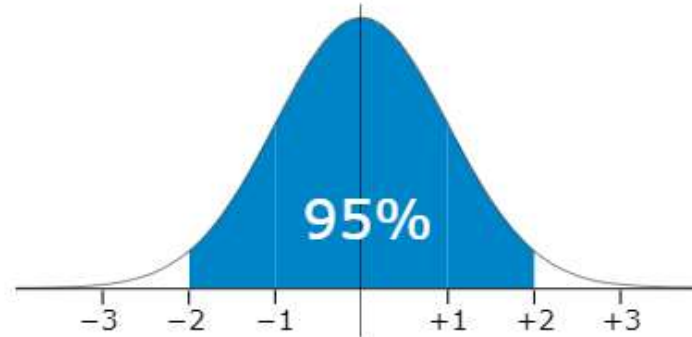
$$X \sim N(\mu, \sigma^2)$$



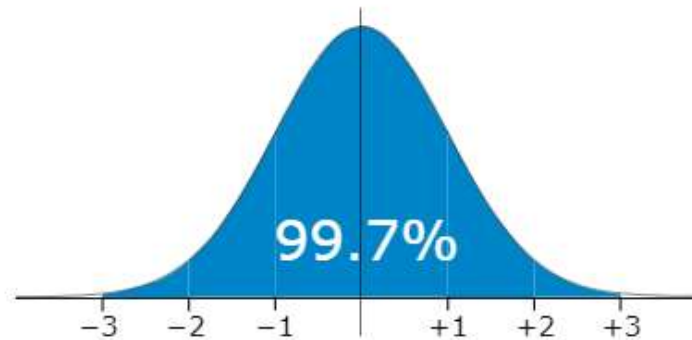
Normal Distribution – Empirical Rule



68% of values are within
1 standard deviation of the mean



95% of values are within
2 standard deviations of the mean



99.7% of values are within
3 standard deviations of the mean

Normal Distribution

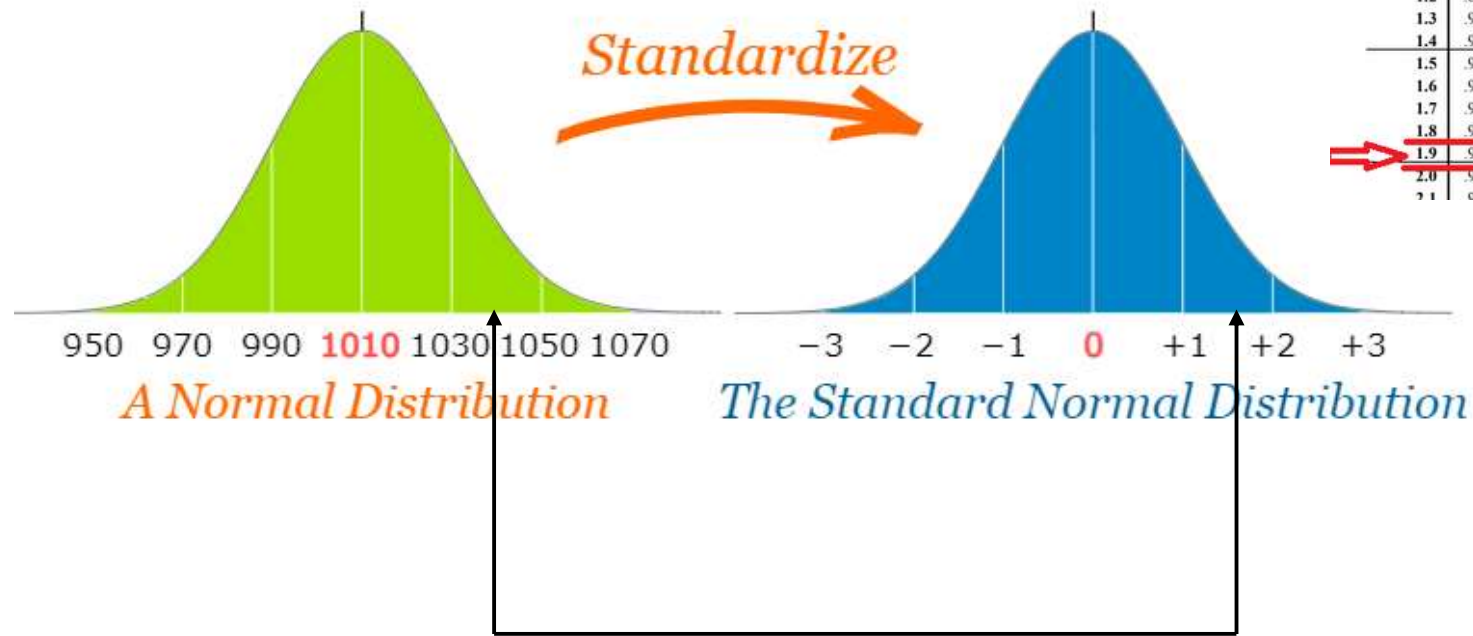
Ex- Mean age of a population is 50. If the standard deviation is 5, what proportion of people are in the population from age 40 to age 60.

$$50 \pm 2 \times 5 = [40, 60]$$

So Empirical Rule says that 95% of the population are in age 40 to 60 range.

What proportion of people are in the population from age 42 to age 58?

Normal Distribution

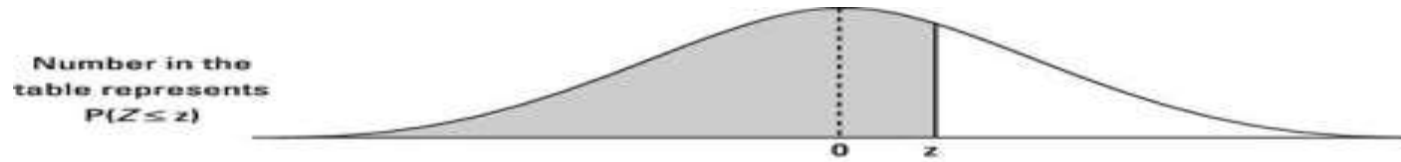


$$z = \frac{x - \mu}{\sigma}$$

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

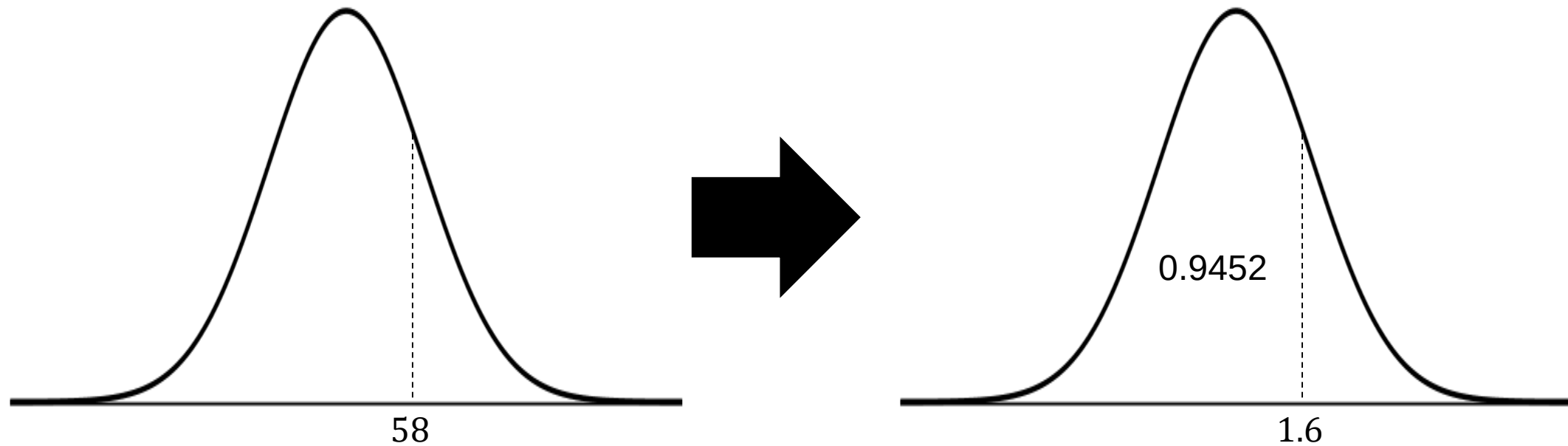
Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574

Normal Distribution – Z Table

[illegible]

Normal Distribution

Assume that the mean is 50 and the standard deviation is 5.



Normal Distribution

What proportion of people are in the population from age 42 to age 58?

- Find the proportion of people below age 58
- Find the proportion of people below age 42
- Get the difference between these proportions

What proportion of people in the population who are older than 58?

Exponential Distribution

- A random variable is defined to have an exponential distribution, if the probability density function of X is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{for } 0 \leq x \leq \infty \end{cases}$$

$$\text{Also, } E(X) = \frac{1}{\lambda} \text{ and } V(X) = \frac{1}{\lambda^2}$$

Exponential Distribution

- Example: At a certain location on a highway, the number of cars exceeding the allowed speed limit per half hours is 8.4. What is the probability of waiting time is 5 minutes?

$$f(X = 5 \text{ minutes}) = 0.28e^{-0.28 \times 5} = 0.07$$

- What is the probability of waiting time less than 5 minutes?

$$f(X < 5 \text{ minutes}) = \int_0^5 0.28e^{-0.28x} dx$$



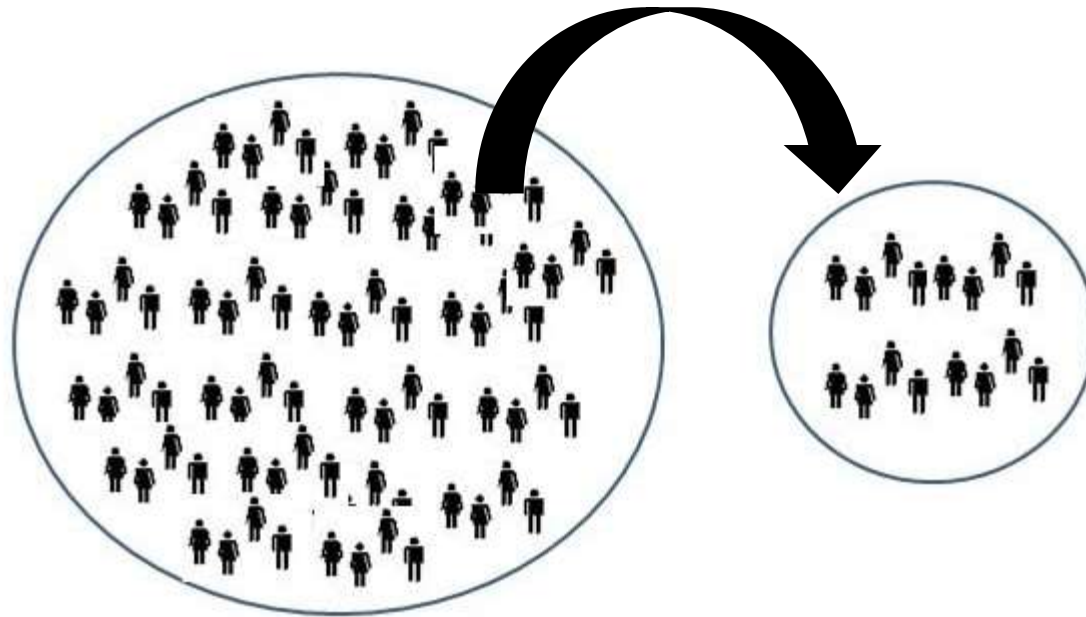
Statistical Inference



H.M. Samadhi Chathuranga Rathnayake

Sample

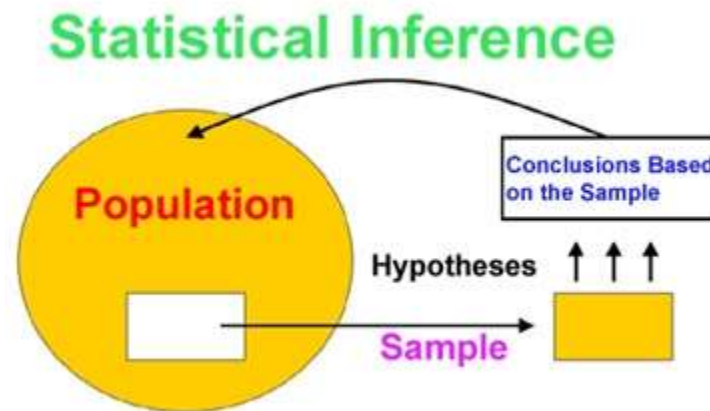
In statistics, a sample is a subset of the population in which a specific matter is going to be checked, or a specific experiment is going to be done.



Statistical Inference

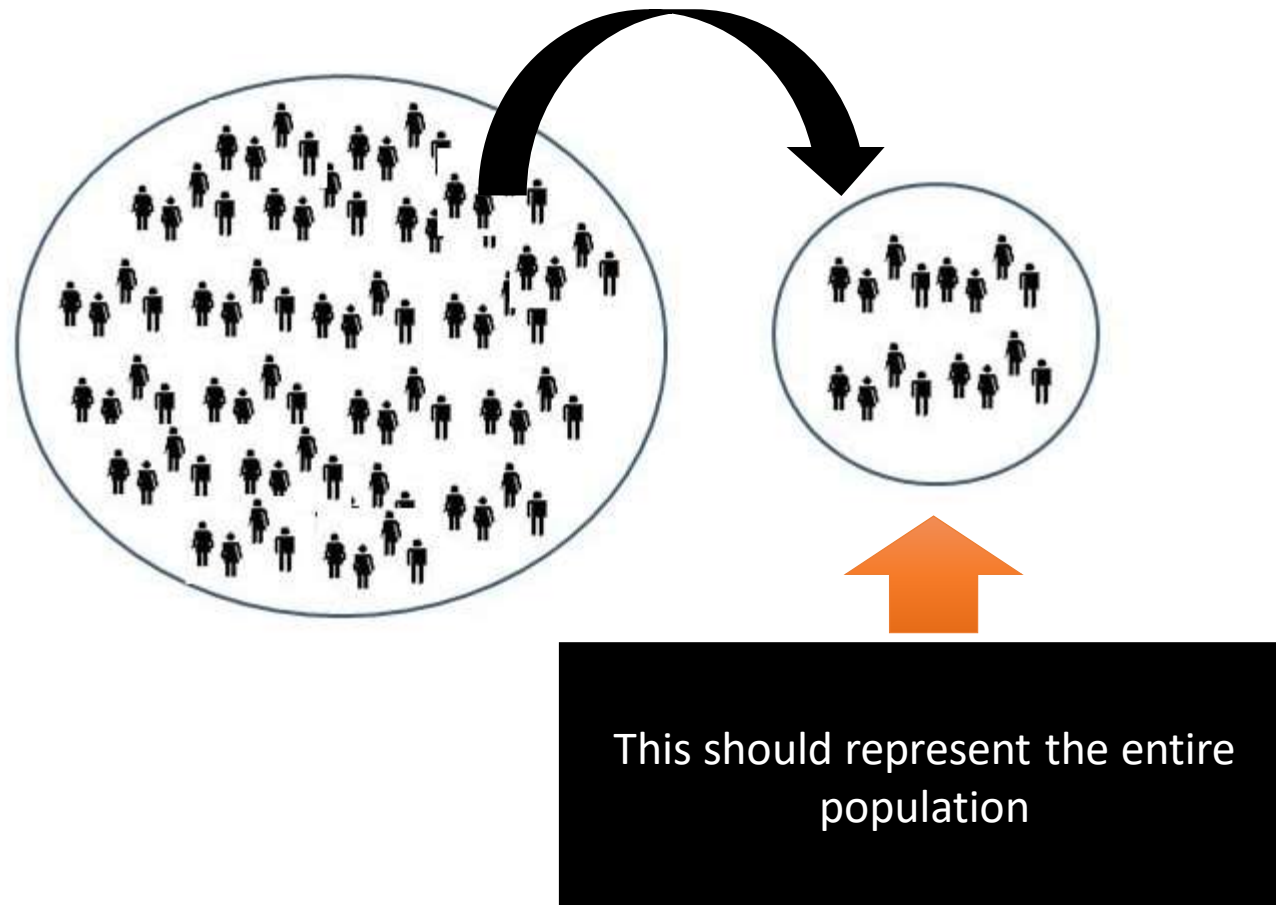
Use data from sample to make estimates, to test claims or hypothesis about the characteristics of the population (Parameters).

The main concern in statistical inference is drawing conclusions about the population not about the sample.



Random Sample

The sample should be a **Random Sample**, where each item in the population has the same probability or chance to be included to the sample.



Sampling

Usually as mentioned above each sample is taken from the population should be a random sample. The process of taking these random samples is specially called as **Probability Sampling**.



But sometimes, because of unavoidable reasons, taking random samples may be difficult to proceed. In those cases, there is another technique called **Non-probability Sampling** can be used. This involves non-random selection based on convenience or other criteria, allowing you to easily collect data.



Probability Sampling

Probability sampling means that every member of the population has a chance of being selected. It is mainly used in quantitative research. If you want to produce results that are representative of the whole population, probability sampling techniques are the most valid choice.

These are normally applied for quantitative researches.

- Simple Random Sampling
- Systematic Sampling
- Stratified Sampling
- Cluster Sampling

Non-Probability Sampling

In a non-probability sample, individuals are selected based on non-random criteria, and not every individual has a chance of being included.

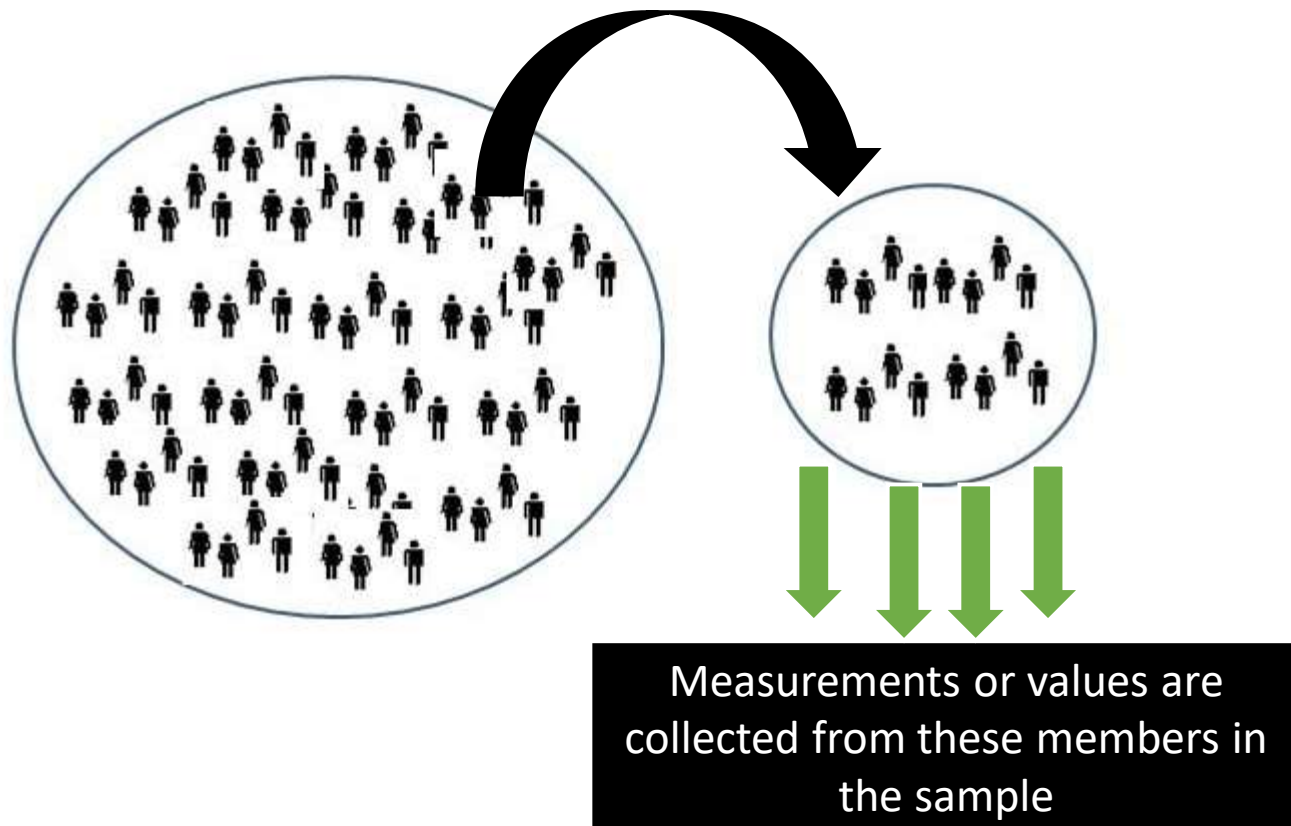
This type of sample is easier and cheaper to access, but it has a higher risk of sampling bias. That means the inferences you can make about the population are weaker than with probability samples, and your conclusions may be more limited. If you use a non-probability sample, you should still aim to make it as representative of the population as possible.

Non-probability sampling techniques are often used in exploratory and qualitative researchs. In these types of research, the aim is not to test a hypothesis about a broad population, but to develop an initial understanding of a small or under-researched population.

- Convenience Sampling
- Voluntary Response Sampling
- Purposive Sampling (Judgement Sampling)
- Snowball Sampling

Sample Values

Since we cannot access to the population, now a sample has been selected and these values will be used to calculate some values for estimating the inaccessible population parameters.



Estimation of Population Parameters

Suppose that the population has an unknown parameter such as mean, variance or the proportion of success. Then an estimate of the unknown parameter can be made from the information supplied by a random sample or samples taken from the population.

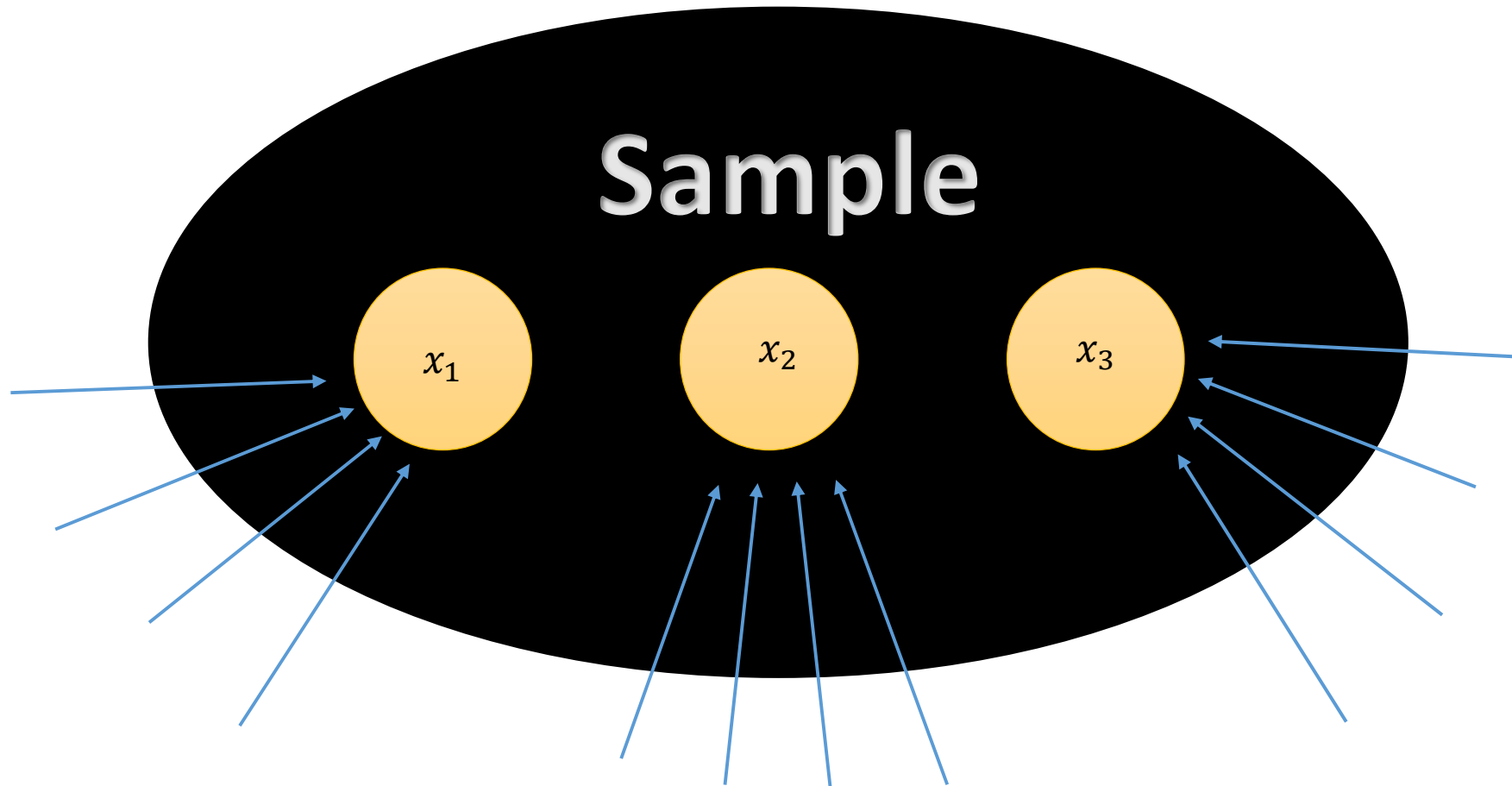
A statistic is used to estimate the value of a parameter is called “Estimator” and denoted using capital letters with hat notations (Ex- $\hat{S}$). The numerical value taken by the estimator is called the estimate.

There are two ways of estimation.

1. Point estimation – Determine an estimator which based on the sample data and will give a single value estimate for an unknown parameter.
2. Interval estimation – Determine an interval that is likely to contain an unknown parameter value.

Sample Statistics

Consider x_1, x_2, x_3 are sample values from a population with the mean μ . These are variables which can take any value from the population.



Sample Statistics

A function or sample values **free** from parameters.

Consider the above x_1, x_2, x_3 sample values. Some possible functions are given below with these variables.

$$F = x_1^2 + x_2^2 + x_3^2$$

Statistic

$$T = \frac{x_1 + x_2 + x_3}{3}$$

Statistic

$$S = x_1 + \mu$$

Not a statistic

Point Estimation

Any statistic can be considered to propose as an estimator. But how to select the correct statistic among the candidates.

Ex- For a population mean parameter any statistic can be proposed as the estimator when x_1 , x_2 and x_3 are random values taken from a population with mean μ and variance σ^2 .

$$F = x_1^2 + x_2^2 + x_3^2$$

$$T = \frac{x_1 + x_2 + x_3}{3}$$

$$U = \frac{2x_1 + 3x_2 + 4x_3}{6} \dots \text{etc}$$

**What is the correct
statistic among
above candidates for
using as the
estimator for the
population mean?**



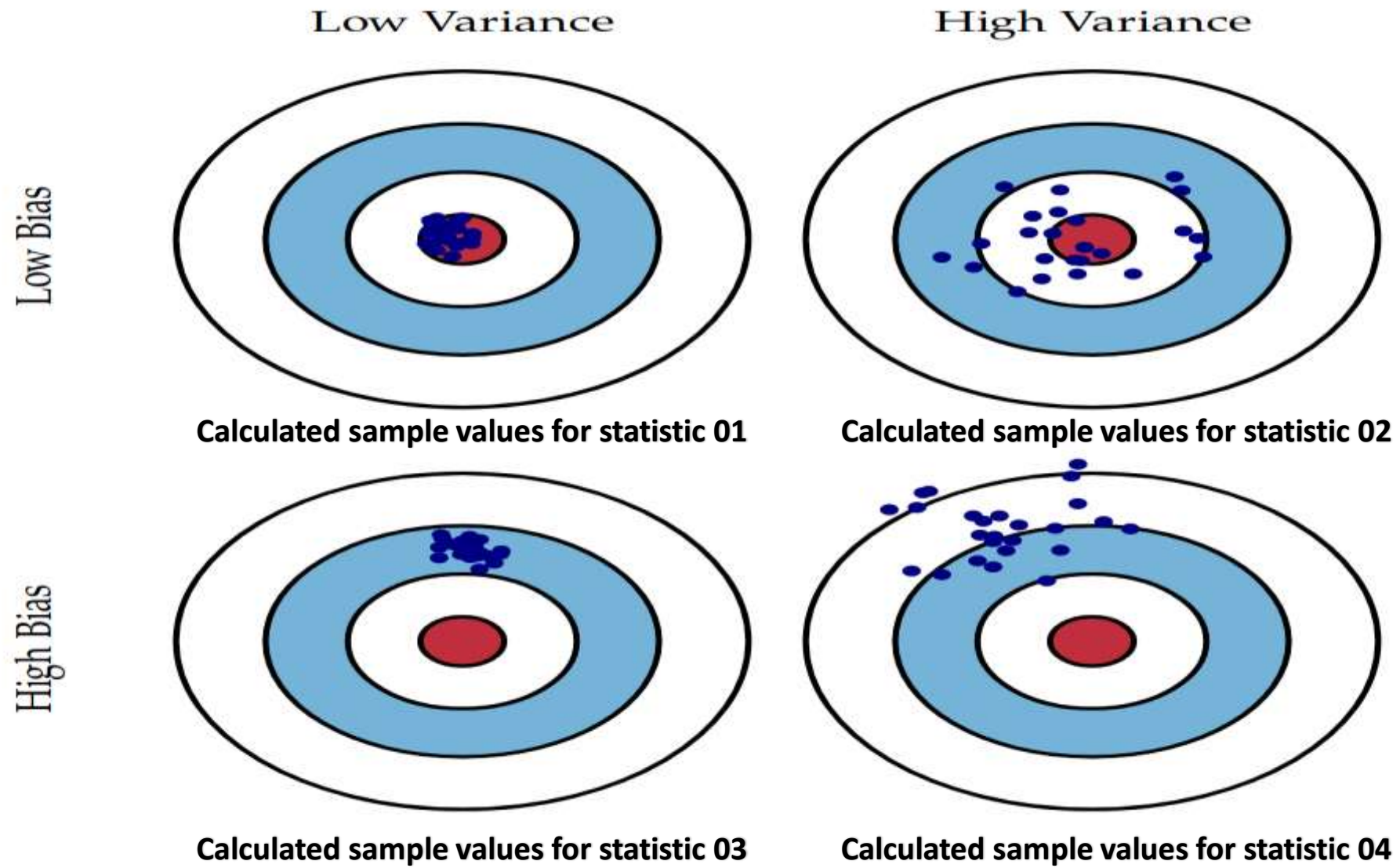
Point Estimation

That will be based on the answers to the following 2 questions.

1. Is it an unbiased estimator?
2. Is it an efficient estimator?

1. Unbiased Estimator – Consider a population with unknown parameter θ . If U is some statistic derived from a random sample taken from the population. Then U is an unbiased estimator for θ if $E(U) = \theta$.
2. Efficient Estimator – One estimator is said to be more efficient than the others when the variability of its sampling distribution is less than the other. ($V(U_1) < V(U_2) \implies U_1$ is more efficient)

What this means?



Unbiased & Most Efficient Estimators for Population Parameters

a) Population mean (μ)

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

a) Population variance (σ^2)

$$\widehat{\sigma^2} = S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$$

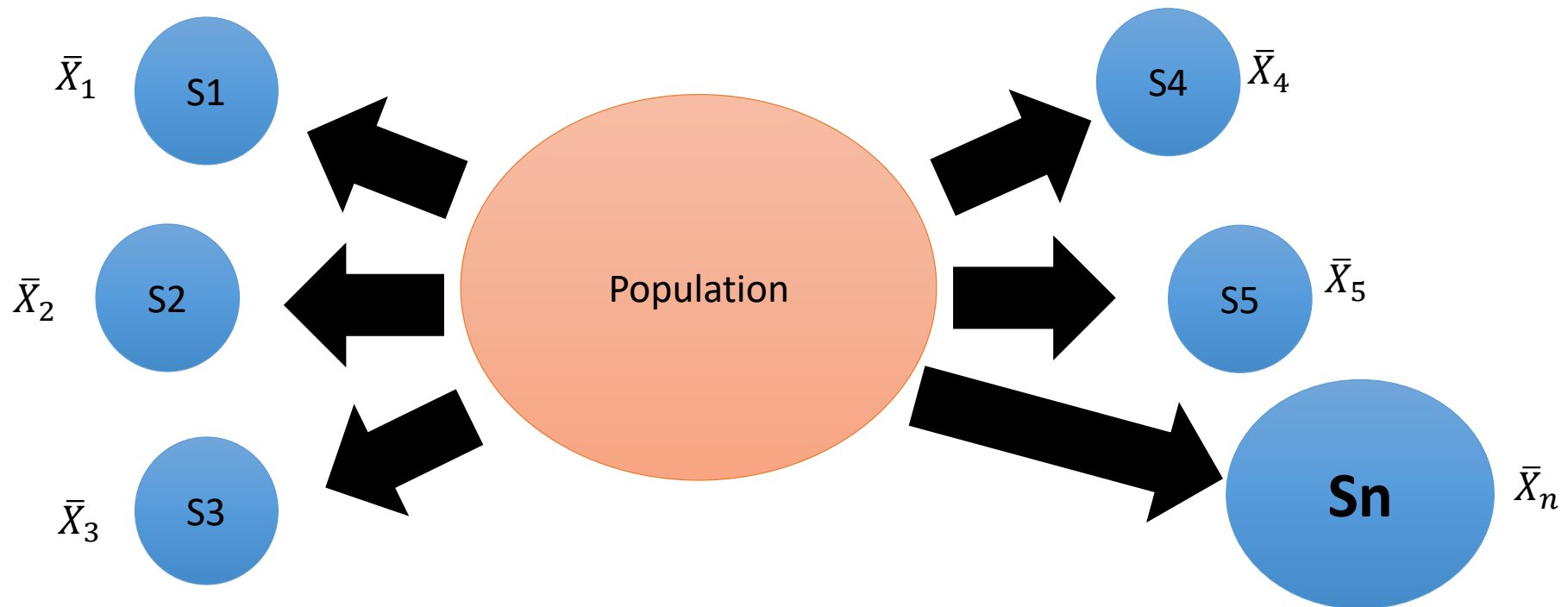
a) Population proportion (P)

$$\hat{P} = \frac{\text{Number of successes in the sample}}{n}$$

Sampling Distribution

In a population any number of samples can be selected. From each sample any statistic can be calculated. All these statistics are estimates for the population parameter interested.

Consider that we want to estimate the population mean of a population. Then we can get many samples from that population and sample means can be calculated.



Sampling Distribution

Sample statistics are random variables. From sample to sample they vary. As a result, sample statistics also have a distribution called the sampling distribution.

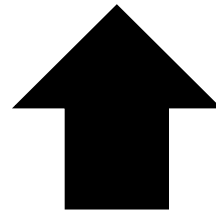
These sampling distributions, similar to distributions discussed previously, have a mean and standard deviation.

Standard deviation of a sampling distribution is called as standard error.

The Distribution of the Sample Mean

If $x_1, x_2, x_3, \dots, x_n$ is a random sample of size n taken from a normal population with mean μ and variance σ^2 , then the distribution of sample mean ($\bar{X}$) is also normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.

Even though the sample is taken from a population which is not distributed normally, when the sample size increases, the sample mean ($\bar{X}$) is also normally distributed with mean μ and variance $\frac{\sigma^2}{n}$.



Central Limit Theorem

The Distribution of the Sample Proportion

Consider a population in which the proportion of success is p . If a random sample of n is taken from this population and then number of successes in the sample (X) allows the binomial distribution.

$$X \sim \text{Bin}(n, p)$$

Now for large n ,

$$X \sim N(np, npq) \quad \text{where } q = 1 - p$$

Now consider $\hat{P} = \frac{X}{n}$. Then,

$$E(\hat{P}) = p \quad \text{and} \quad V(\hat{P}) = \frac{pq}{n}$$

So, for large samples ($n > 30$),

$$\hat{P} \sim N\left(p, \frac{pq}{n}\right)$$

Interval Estimation

The purpose of using an interval estimator is to have some degree of confidence of securing the true parameter. For an interval estimator of a single parameter θ , we will use the random sample to find two quantities L and U such that $L \leq \theta \leq U$. This interval (L, U) should have two properties:

- $P(L \leq \theta \leq U)$ is high.
- The length of the interval (L, U) should be relatively narrow on average.

Interval estimators are called **confidence intervals** and the limits are called **the upper and lower confidence limits**. The probability that a confidence interval will contain the true parameter θ is called the **confidence coefficient** or **confidence level**. This gives the fraction of the time that the constructed interval will contain the true parameter, under repeated sampling.

Interval Estimation

Let L and U be the lower and upper limits for a parameter θ based on a random sample $x_1, x_2, x_3, \dots, x_n$. Both L and U are functions of the sample.

We can write the interval estimate of θ as:

$$P(L \leq \theta \leq U) = 1 - \alpha$$

- It is $(1 - \alpha)100\%$ confident that the true parameter θ is located in the interval (L, U) .
- (L, U) is referred to as a $(1 - \alpha)100\%$ confidence interval for θ .
- In simple words, **when we create infinite intervals with the $(1 - \alpha)100\%$ confidence by getting random samples of the population repeatedly, $(1 - \alpha)100\%$ of them are containing the true parameter θ .**



Hypothesis Testing



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Hypothesis Testing

When making a statistical inquiry, we often put forward a claim (hypothesis) about a population parameter.

Ex-

- Mean height of 15 year old boys is 1.62m
- The proportion of diabetes in a particular district is 0.05
- The mean number of telephone calls received by a telephone operator over a period of 60 minutes is 20.

The hypothesis we put forward is called the **Null Hypothesis** and is denoted by H_0 . Just define Null Hypotheses for above examples as below.

- $H_0 : \mu = 1.62$
- $H_0 : p = 0.05$
- $H_0 : \mu = 20$

In order to test the validity of H_0 we consider observations made on the random sample taken from the populations and perform a statistical test.

Hypothesis Testing

This test shows that whether we should reject the null hypothesis (rejecting the claim stated under H_0), we do so in favor of an **Alternative Hypothesis**, which is usually the opposite of what is stated in the null hypothesis and is denoted by H_1 or H_a .

Defining Hypotheses

When you get a question, an easy way to define hypotheses is, putting the question under the alternative hypothesis.

Ex-

1. The UGC Sri Lanka needs to know that whether the “majority of students at University of Moratuwa are from Western Province”

$$H_1 : p > 0.5 \text{ and } H_0 : p \leq 0.5 \longrightarrow H_0 : p = 0.5$$

2. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42000 miles as the tire manufacturer advertises that the average lifetime is at least 42000.
3. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.
4. A political news company believes the national approval rating for the current president has fallen below 40%.

Defining Hypotheses

2. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42000 miles as the tire manufacturer advertises that the average lifetime is at least 42000.

$$H_1 : \mu < 42000 \text{ and } H_0 : \mu = 42000$$

3. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.

$$H_1 : \mu \neq 8.5 \text{ and } H_0 : \mu = 8.5$$

4. A political news company believes the national approval rating for the current president has fallen below 40%.

$$H_1 : p < 0.4 \text{ and } H_0 : p = 0.4$$

Type I Versus Type II Errors

When testing hypotheses there are essential two errors that can be made.

1. **Type I error** - Rejecting the null hypothesis when the null is correct actually.
2. **Type II error** - Not rejecting the null hypothesis when the null is not correct actually.

Decision	Reality	
	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct
Do not reject H_0	Correct	Type II error

$$P(\text{Type I error}) = \alpha$$

$$P(\text{Type II error}) = \beta$$

Significance Level

When we conduct a hypothesis test we will always have to specify a significance level. We basically have to answer the question – “What is the probability of a Type I error we are willing to live with?”.

We cannot generally quantify the probability of a type II error. To avoid making Type II errors we will always say “Do not reject the null hypothesis” instead of “Accept the null hypothesis”.





One Population Mean



H.M. Samadhi Chathuranga Rathnayake

Chi – square Distribution

Chi-square distributions are used in many inferential problems, for example, in inferential problems dealing with variance. It is a probability distribution which consists of a single parameter. If n is a positive integer, then the parameter is called the degrees of freedom. The mean & variance of a Chi-square distribution are n and $2n$ respectively.

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$. Then we know that,

$$Z_i = \frac{(X_i - \mu)}{\sigma} \sim N(0,1)$$

Then

$$Z_i^2 \sim \chi_{(1)}^2$$

Then

$$\sum_{i=1}^n Z_i^2 \sim \chi_{(n)}^2$$

Chi – square Distribution

Chi-square distributions are used in many inferential problems, for example, in inferential problems dealing Let $x_1, x_2, x_3, \dots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$. The we know that,

1. When μ is known,

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2_{(n)}$$

2. When μ is unknown,

$$\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \sim \chi^2_{(n-1)}$$

Since $S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

t Distribution

If Y and Z are independent random variables, Y has a Chi-square distribution with n degrees of freedom and $Z \sim N(0,1)$, then

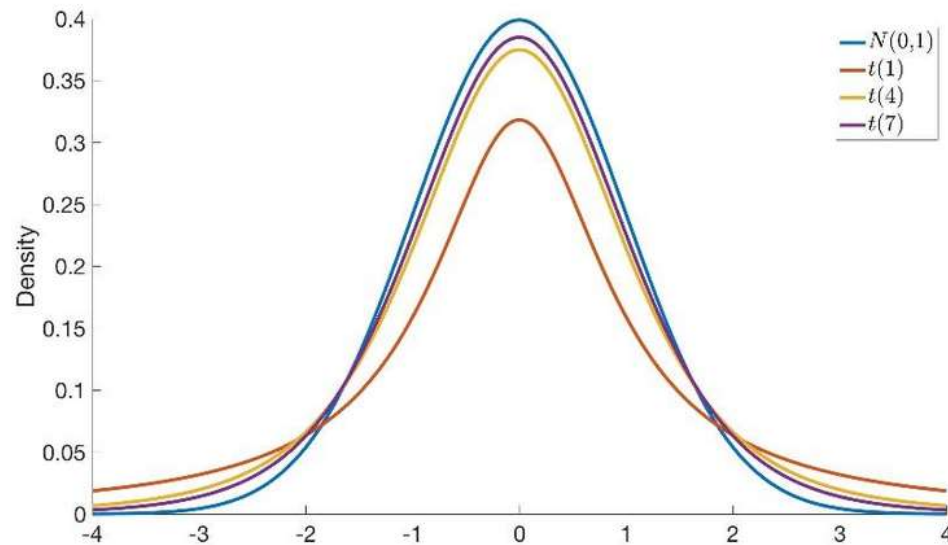
$$T = \frac{Z}{\sqrt{\frac{Y}{n}}}$$

is said to have a t distribution with n degrees of freedom. That is denoted by $T \sim t_{(n)}$.

t Distribution

The t distribution is similar to the standard normal distribution, but the height of the t distribution differs depending on the sample size. **t distribution is used when we don't know the population standard deviation and when we have small sample sizes (Say $n < 30$).** When the sample size increases, the t distribution approaches the standard normal distribution.

The height of the t distribution is determined by the number of degrees of freedom.



t Distribution

Sample mean and sample variance are independent.

$$Z = \frac{(\bar{X} - \mu)}{\sigma / \sqrt{n}} \sim N(0,1)$$

We know that,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

So

$$\left(\frac{(\bar{X} - \mu)}{\sigma / \sqrt{n}} \right) / \sqrt{\frac{(n-1)S^2}{\sigma^2} \cdot \frac{1}{n-1}} = \frac{(\bar{X} - \mu)}{S / \sqrt{n}} \sim t_{(n-1)}$$

F Distribution

Let U and V be Chi-squared random variables with n_1 and n_2 degrees of freedom respectively. If U and V are independent.

$$F = \frac{U/n_1}{V/n_2} \sim F(n_1, n_2)$$

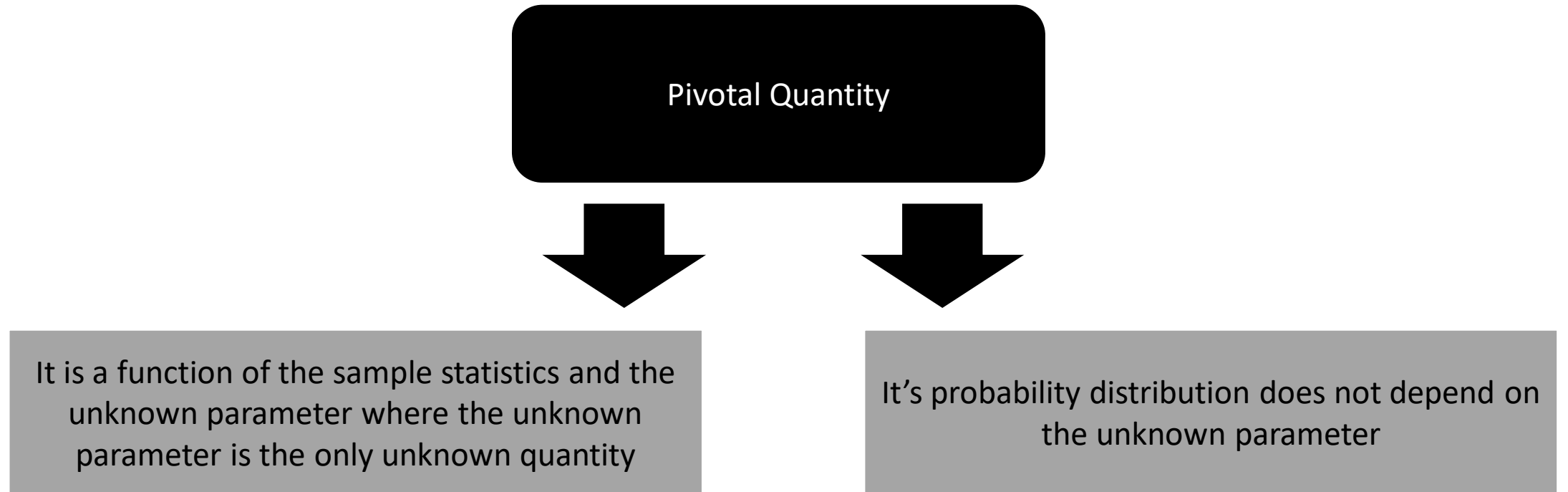
Now consider two different populations with different population variances.

$$\frac{(n_1 - 1)S_1^2}{\sigma_1^2} \sim \chi_{(n_1-1)}^2 \text{ and } \frac{(n_2 - 1)S_2^2}{\sigma_2^2} \sim \chi_{(n_2-1)}^2$$

So,

$$\frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F(n_1 - 1, n_2 - 1)$$

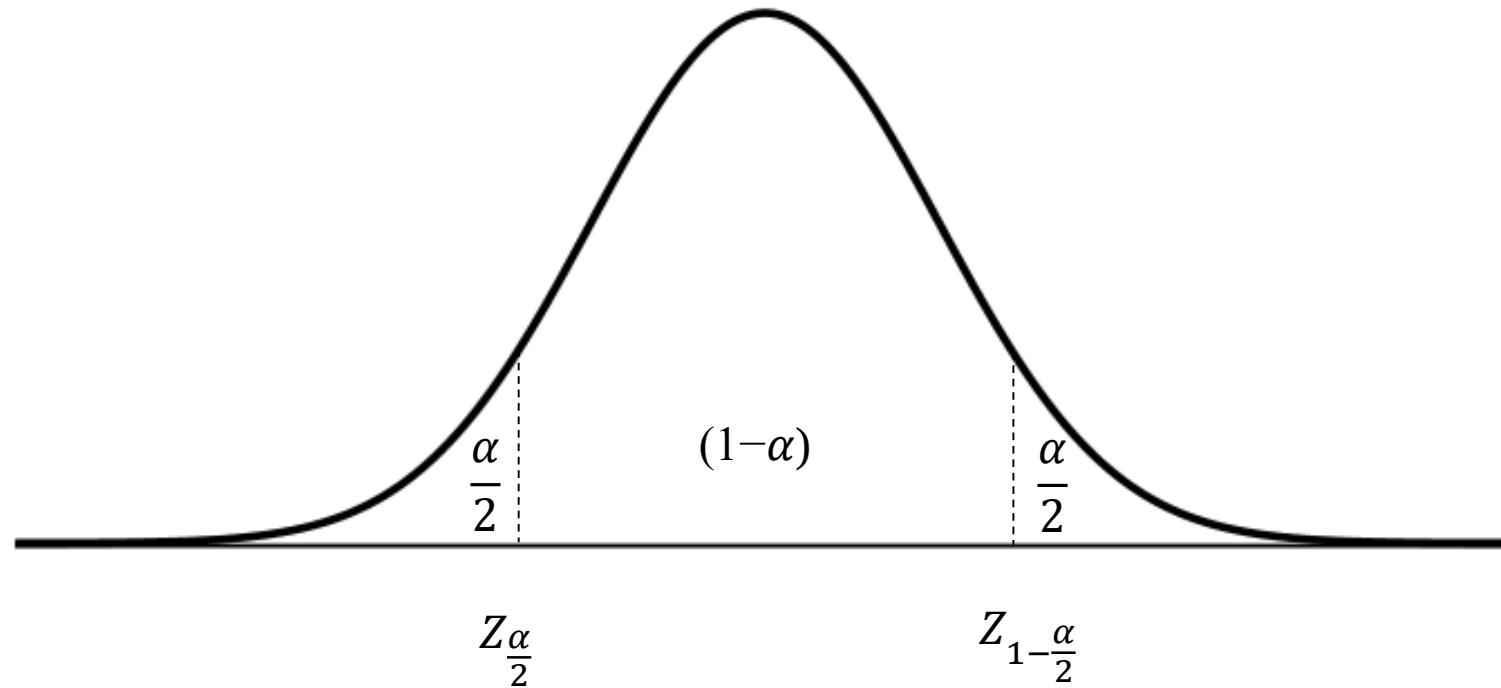
The Pivotal Method of Finding the Confidence Interval



$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is known

In a random sample $x_1, x_2, x_3, \dots, x_n$ from $N(\mu, \sigma^2)$, $\bar{X} \sim N(\mu, \sigma^2/n)$.

By standardizing $\bar{X}$, $Z = \frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \sim N(0,1)$



$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is known

We have to create,

$$P(L \leq \mu \leq U) = 1 - \alpha$$

For creating that we use the pivotal quantity: $Z = \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}}$

So,

$$P\left(Z_{\frac{\alpha}{2}} \leq \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}} \leq Z_{(1-\frac{\alpha}{2})}\right) = P\left(-Z_{(1-\frac{\alpha}{2})} \leq \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}} \leq Z_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha$$

By simplifying,

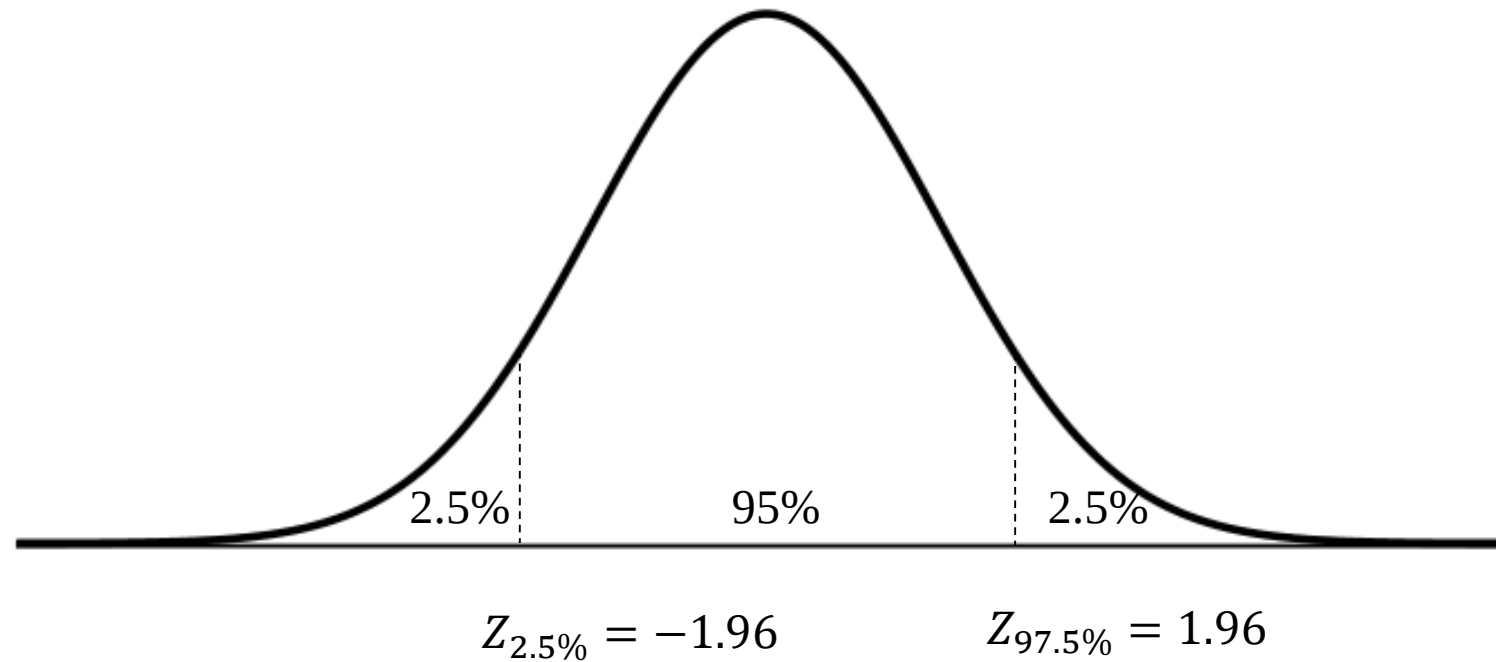
$$P\left(\bar{X} - \sigma/\sqrt{n} Z_{(1-\frac{\alpha}{2})} \leq \mu \leq \bar{X} + \sigma/\sqrt{n} Z_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha$$

Therefore a $(1-\alpha)100\%$ confidence interval for μ is:

$$\left[\bar{X} - \sigma/\sqrt{n} Z_{(1-\frac{\alpha}{2})} \quad , \quad \bar{X} + \sigma/\sqrt{n} Z_{(1-\frac{\alpha}{2})}\right]$$

Ex-

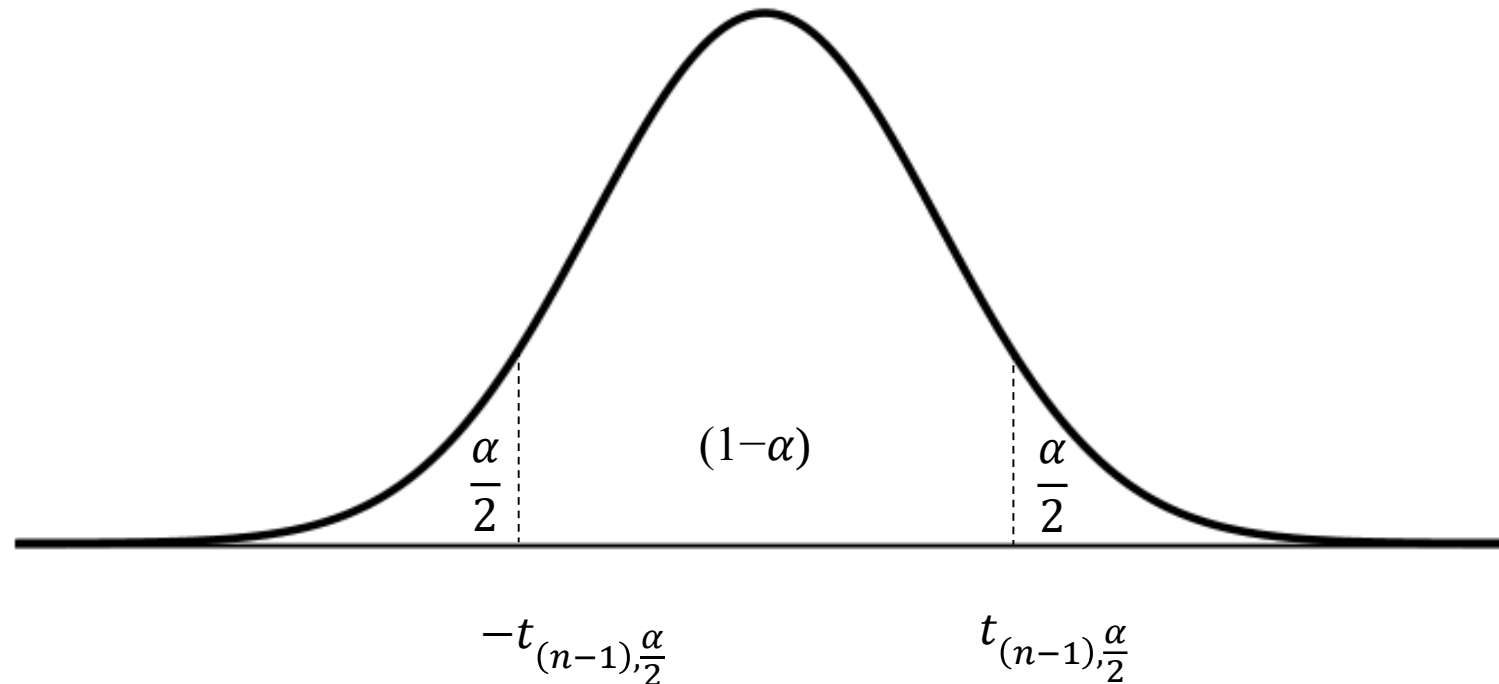
A random sample of 12 items is selected from a normal population with variance 4. Sample mean is 10.4. Find the 95% confidence interval for the population mean.



$$\left[10.4 - \frac{2}{\sqrt{12}} Z_{2.5\%}, 10.4 + \frac{2}{\sqrt{12}} Z_{2.5\%} \right] = [9.26, 11.52]$$

$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is unknown

Suppose we have a random sample $x_1, x_2, x_3, \dots, x_n$ from $N(\mu, \sigma^2)$ where σ^2 is unknown. Let's focus about t distribution. Consider $T = \frac{(\bar{X}-\mu)}{s/\sqrt{n}} \sim t_{(n-1)}$



$(1-\alpha)100\%$ Confidence Interval for mean (μ) when variance is unknown

We have to create,

$$P(L \leq \mu \leq U) = 1 - \alpha$$

For creating that we use the pivotal quantity: $T = \frac{(\bar{X} - \mu)}{S/\sqrt{n}} \sim t_{(n-1)}$

So,

$$P\left(-t_{(n-1), \frac{\alpha}{2}} \leq \frac{(\bar{X} - \mu)}{S/\sqrt{n}} \leq t_{(n-1), \frac{\alpha}{2}}\right) = 1 - \alpha$$

By simplifying,

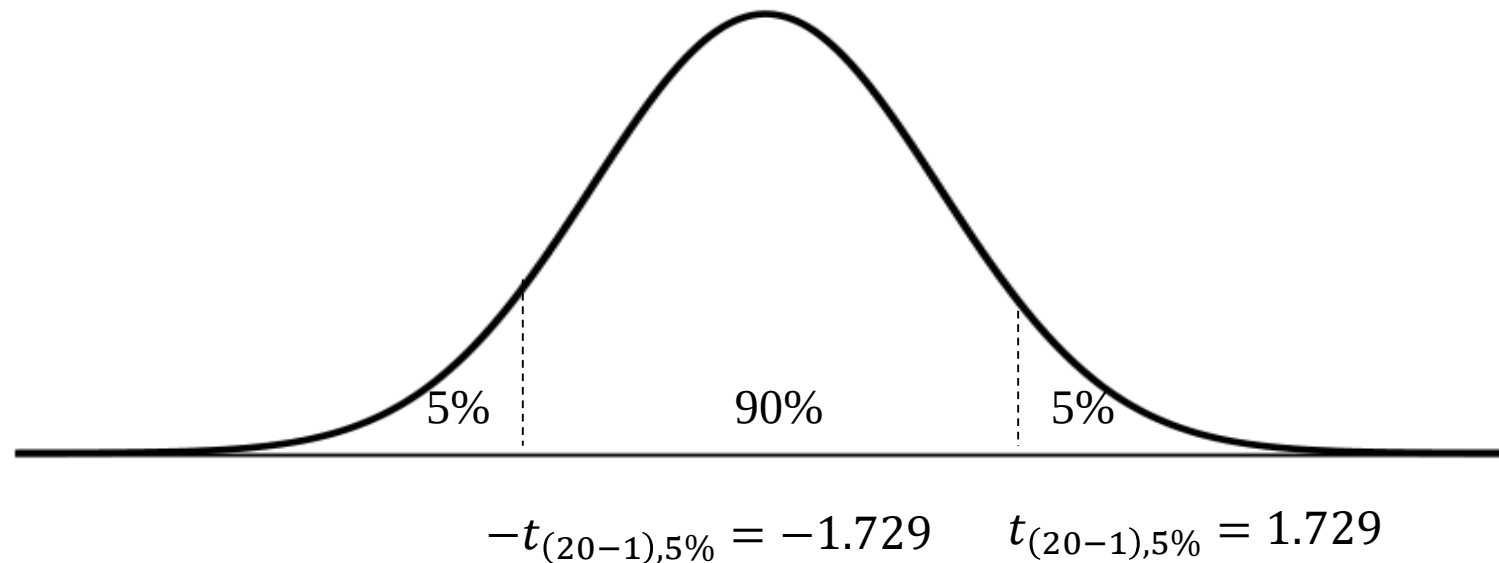
$$P\left(\bar{X} - S/\sqrt{n} t_{(n-1), \frac{\alpha}{2}} \leq \mu \leq \bar{X} + S/\sqrt{n} t_{(n-1), \frac{\alpha}{2}}\right) = 1 - \alpha$$

Therefore a $(1-\alpha)100\%$ confidence interval for μ is:

$$\left[\bar{X} - S/\sqrt{n} t_{(n-1), \frac{\alpha}{2}} \quad , \quad \bar{X} + S/\sqrt{n} t_{(n-1), \frac{\alpha}{2}} \right]$$

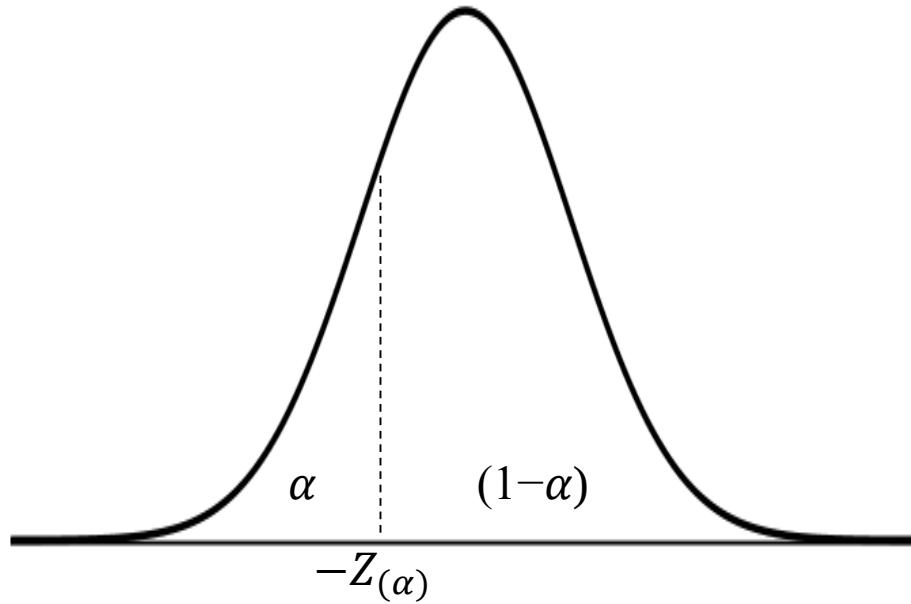
Ex-

Let X equal the amount of butterfat in pounds produced by a typical cow during a 105-day milk production period between her first and second calves. Assume that the distribution of X is normal but the population variance is unknown. Construct a 90% confidence interval for population mean when the sample mean is 507.7 and sample standard deviation is 89.75. (n=20)



$$\left[507.5 - 89.75/\sqrt{20} t_{(20-1),5\%}, 507.5 + 89.75/\sqrt{20} t_{(20-1),5\%} \right] = [472.801, 542.199]$$

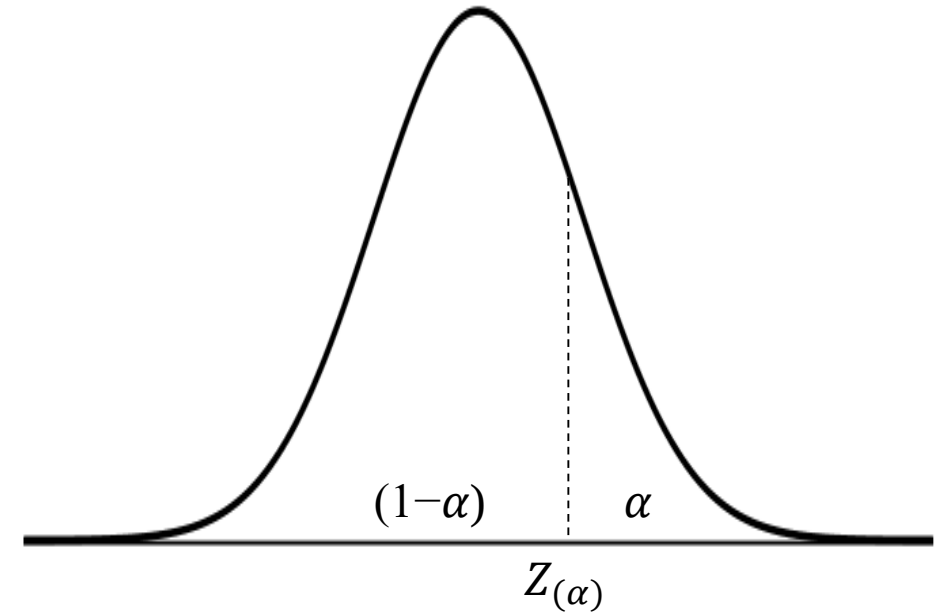
One Sided Confidence Interval for mean (μ) when variance is known



$$P\left(\frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \geq -Z_{(\alpha)}\right) = 1 - \alpha$$

$$P\left(\mu \leq \bar{X} + \sigma/\sqrt{n} Z_{(\alpha)}\right) = 1 - \alpha$$

$$\left[-\infty, \quad \bar{X} + \sigma/\sqrt{n} Z_{(\alpha)}\right]$$



$$P\left(\frac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \leq Z_{(\alpha)}\right) = 1 - \alpha$$

$$P\left(\mu \geq \bar{X} - \sigma/\sqrt{n} Z_{(\alpha)}\right) = 1 - \alpha$$

$$\left[\bar{X} - \sigma/\sqrt{n} Z_{(\alpha)}, \infty\right]$$

Type I Versus Type II Errors

When testing hypotheses there are essential two errors that can be made.

1. **Type I error** - Rejecting the null hypothesis when the null is correct actually.
2. **Type II error** - Not rejecting the null hypothesis when the null is not correct actually.

Decision	Reality	
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Reject H_0	Type I error	Correct
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$$P(\text{Type I error}) = \alpha$$

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Significance Level

When we conduct a hypothesis test we will always have to specify a significance level. We basically have to answer the question – “What is the probability of a Type I error we are willing to live with?”.

We cannot generally quantify the probability of a type II error. To avoid making Type II errors we will always say “Do not reject the null hypothesis” instead of “Accept the null hypothesis”.



Perform Hypothesis Testing

There are three approaches to the hypothesis testing.

(1) Classical Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Find the appropriate test statistic and its distribution.
- Compute the test statistic under H_0 .
- Define the rejection region according to the given significance level.
- If the calculated test statistic falls in the rejection region, reject H_0 and if not, do not reject H_0 .

Perform Hypothesis Testing

There are three approaches to the hypothesis testing.

(2) Confidence Interval Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Construct appropriate confidence interval for specified level of significance under H_0 .
- If the value of the parameter under H_0 lies within the confidence interval, we do not reject H_0 . If the value of the parameter under H_0 lies outside the confidence interval we reject H_0 .

Perform Hypothesis Testing

There are three approaches to the hypothesis testing.

(3) P-Value Approach:

- Define null and alternative hypothesis.
- Check whether the hypothesis is one tailed or two tailed.
- Find the appropriate test statistic and its distribution.
- Compute the P-value based on computed value of the test statistic under H_0 .
- If the P-value is less than the significance level, we reject H_0 . Otherwise we do not reject H_0 .

Tests About Mean : Z – test when population variance is known

Consider the following hypothesis for the population mean, suppose to be tested using a random sample of size n . **This is a two tailed test.**

$$H_0 : \mu = A$$

$$H_1 : \mu \neq A$$

Get a sample from the population and calculate the sample mean $\bar{X}$.

Now consider the test statistic,

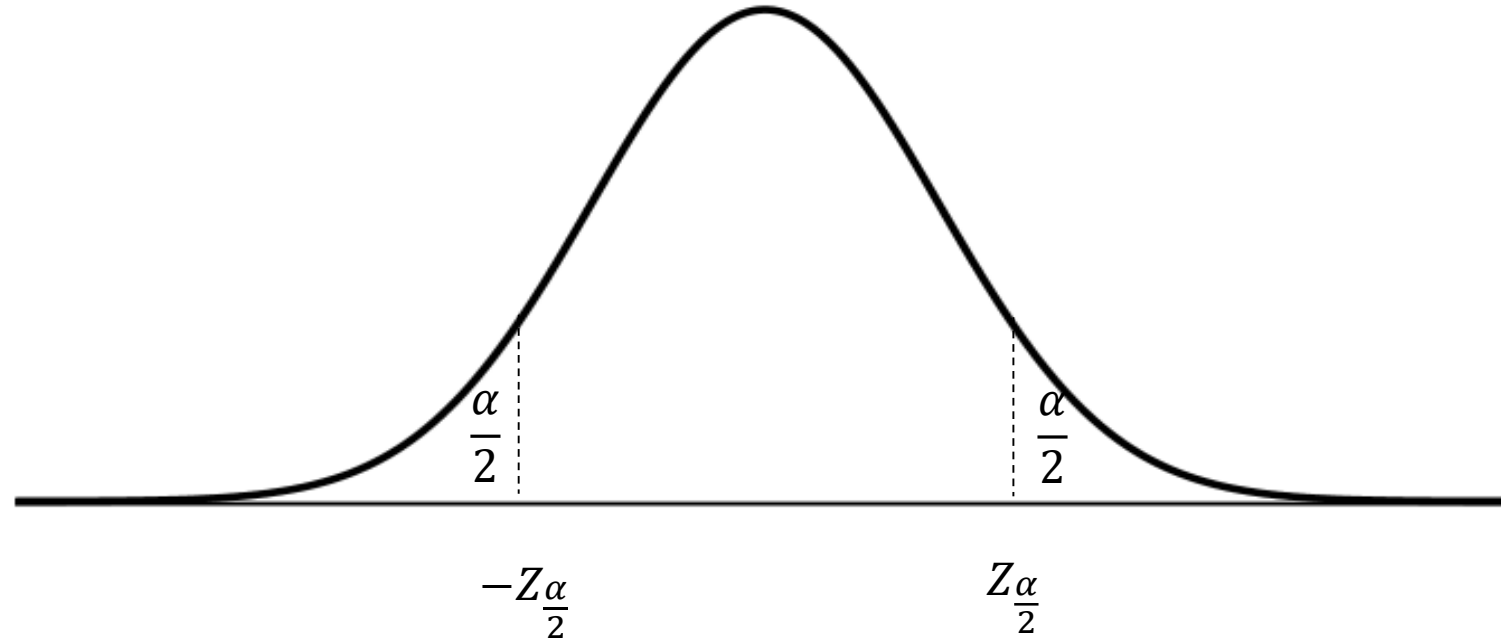
$$Z = \frac{(\bar{X} - \mu)}{\sigma / \sqrt{n}} \sim N(0,1)$$

Now compute the values of the test statistic under H_0 . Then,

$$Z_{cal} = \frac{(\bar{X} - A)}{\sigma / \sqrt{n}}$$

Tests About Mean : Z – test when population variance is known

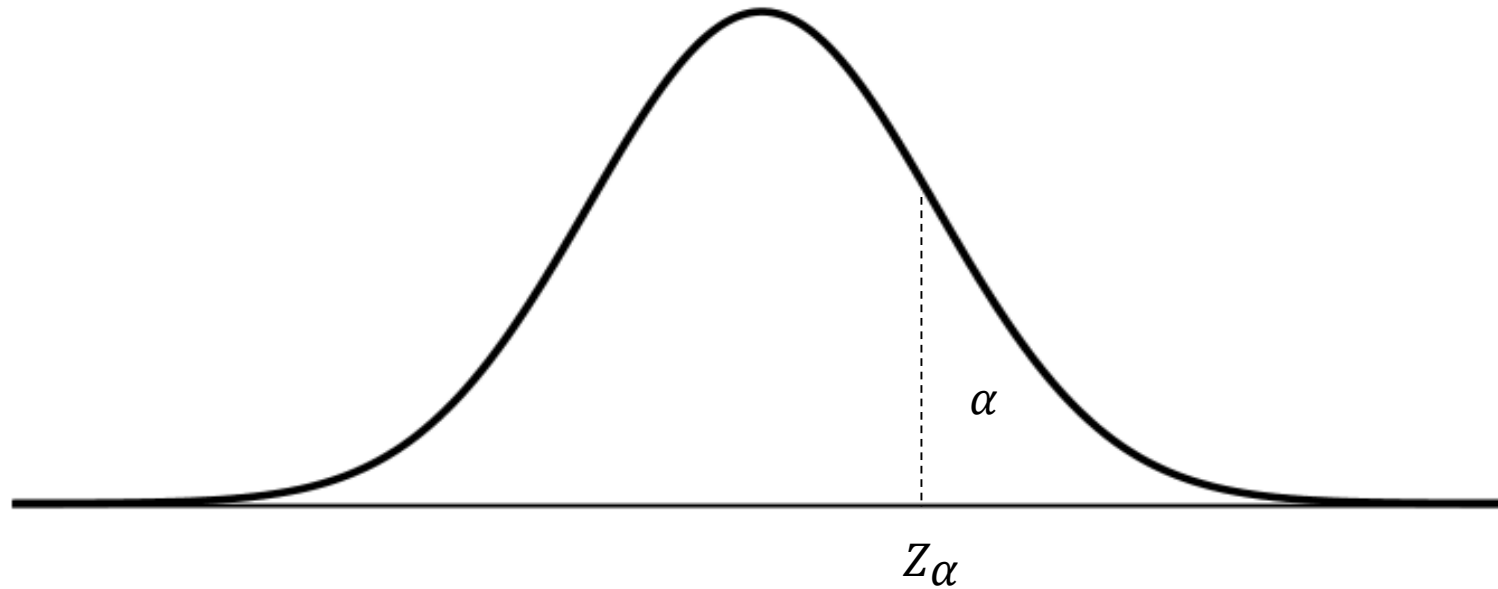
Since this is a two tail test,



If $Z_{cal} > Z_{\frac{\alpha}{2}}$ or $Z_{cal} < -Z_{\frac{\alpha}{2}}$ we reject H_0

Tests About Mean : Z – test when population variance is known

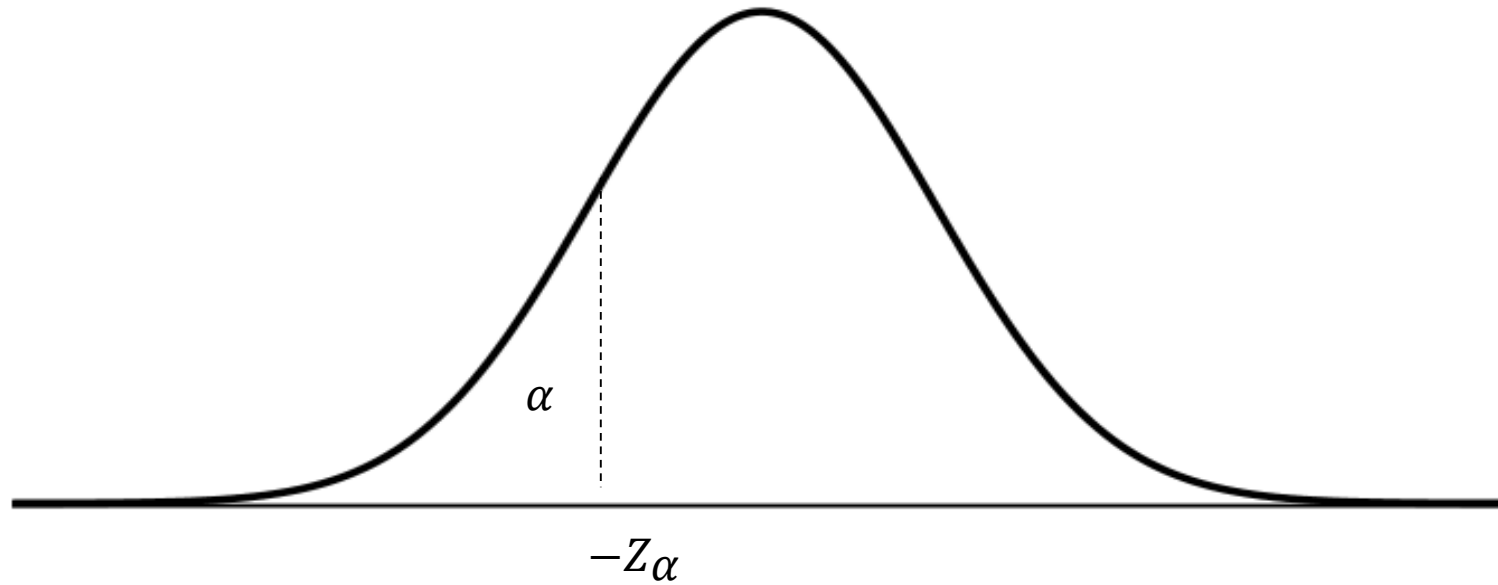
In a one tail – right tail test,, $H_0 : \mu = A$ and $H_1 : \mu > A$



If $Z_{cal} > Z_\alpha$ we reject H_0

Tests About Mean : Z – test when population variance is known

In a one tail – left tail test,, $H_0 : \mu = A$ and $H_1 : \mu < A$



If $Z_{cal} < -Z_\alpha$ we reject H_0

Ex-

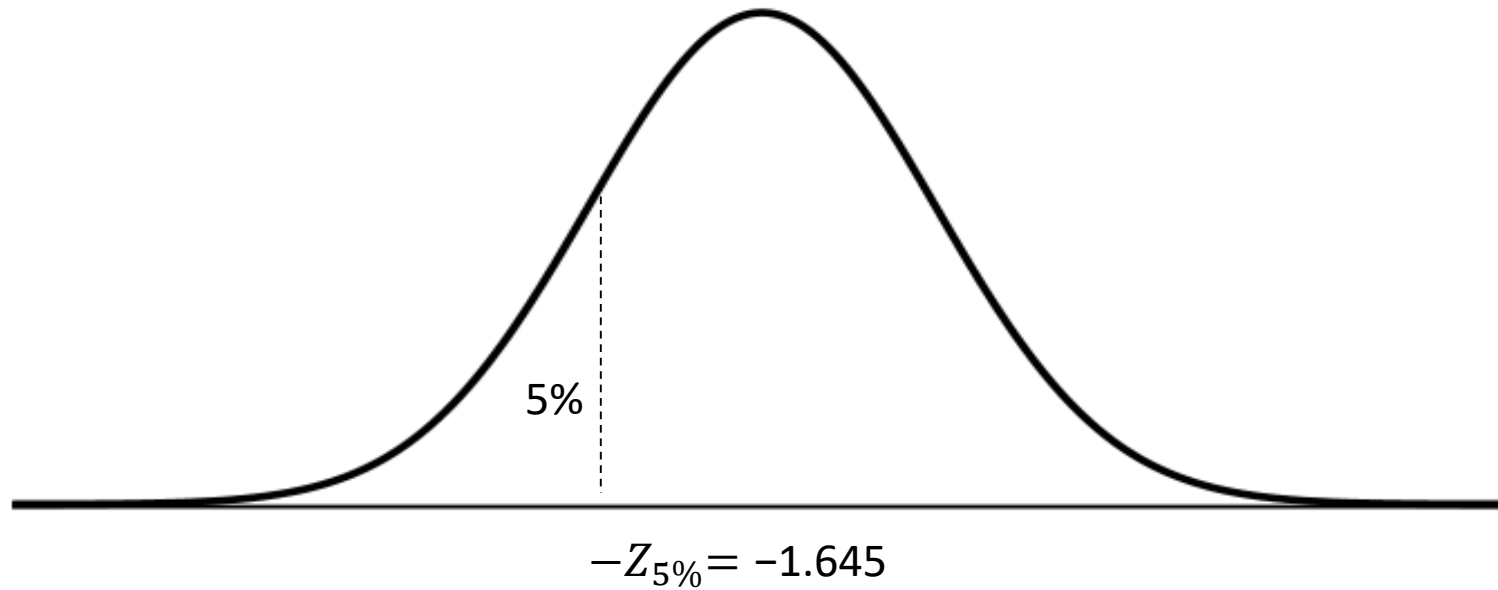
Boys of a certain age are known to have a mean weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. 25 boys from the home are weighed and found to have a mean weight of 80.94 pounds. It is known that the population standard deviation is 11.6 pounds. Based on the available data what should be concluded concerning the complaint? (Significance level 5%)

$$H_0 : \mu = 85$$

$$H_1 : \mu < 85$$

Under H_0 ,

$$Z_{cal} = \frac{(80.94 - 85)}{11.6 / \sqrt{25}} = -1.75$$



$Z_{cal} = -1.75 < -Z_{5\%}$, so we reject H_0 at 5% level of significance.

Conclusion – There is an evidence to say that, the boys living in a municipal children’s home are underfed at 5% level of significance.

[illegible]

Tests About Mean : t – test when population variance is unknown

Consider the following hypothesis for the population mean, suppose to be tested using a random sample of size n . **This is a two tailed test.**

$$H_0 : \mu = A$$

$$H_1 : \mu \neq A$$

Get a sample from the population and calculate the sample mean $\bar{X}$.

Now consider the test statistic,

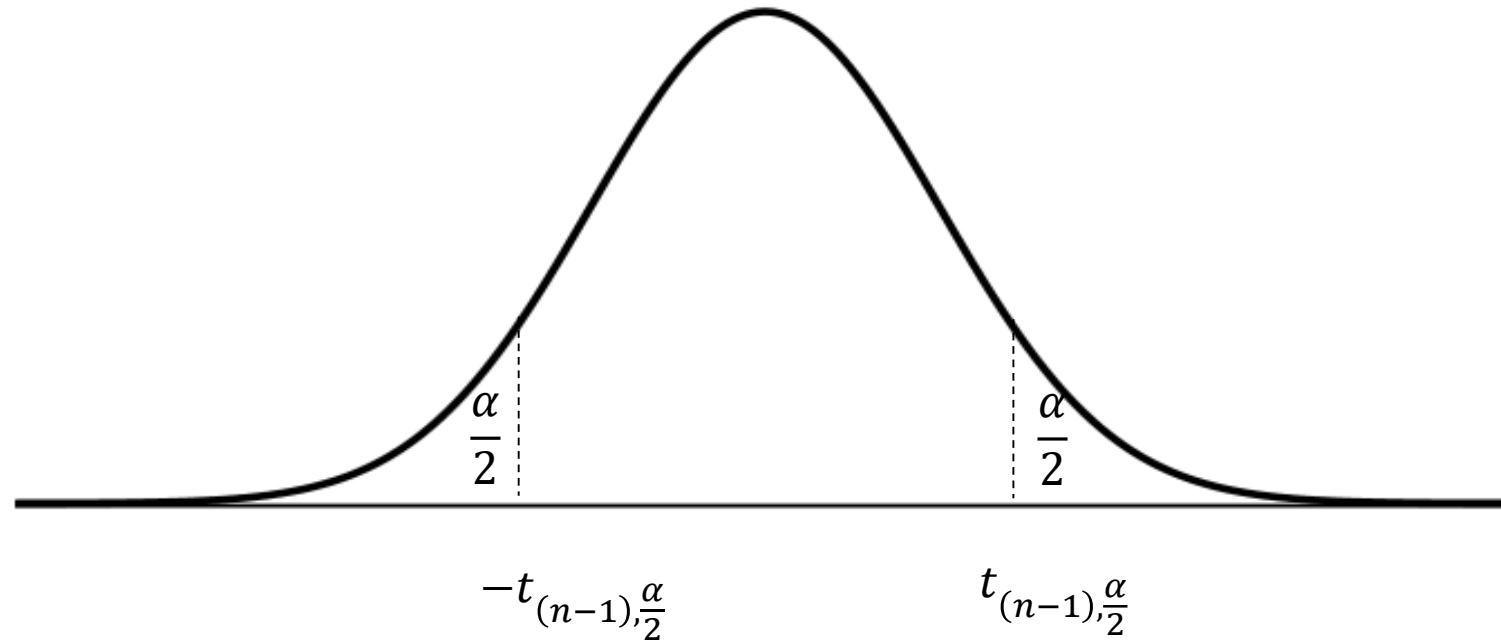
$$T = \frac{(\bar{X} - \mu)}{S/\sqrt{n}} \sim t_{(n-1)}$$

Now compute the values of the test statistic under H_0 . Then,

$$t_{cal} = \frac{(\bar{X} - A)}{S/\sqrt{n}}$$

Tests About Mean : t – test when population variance is unknown

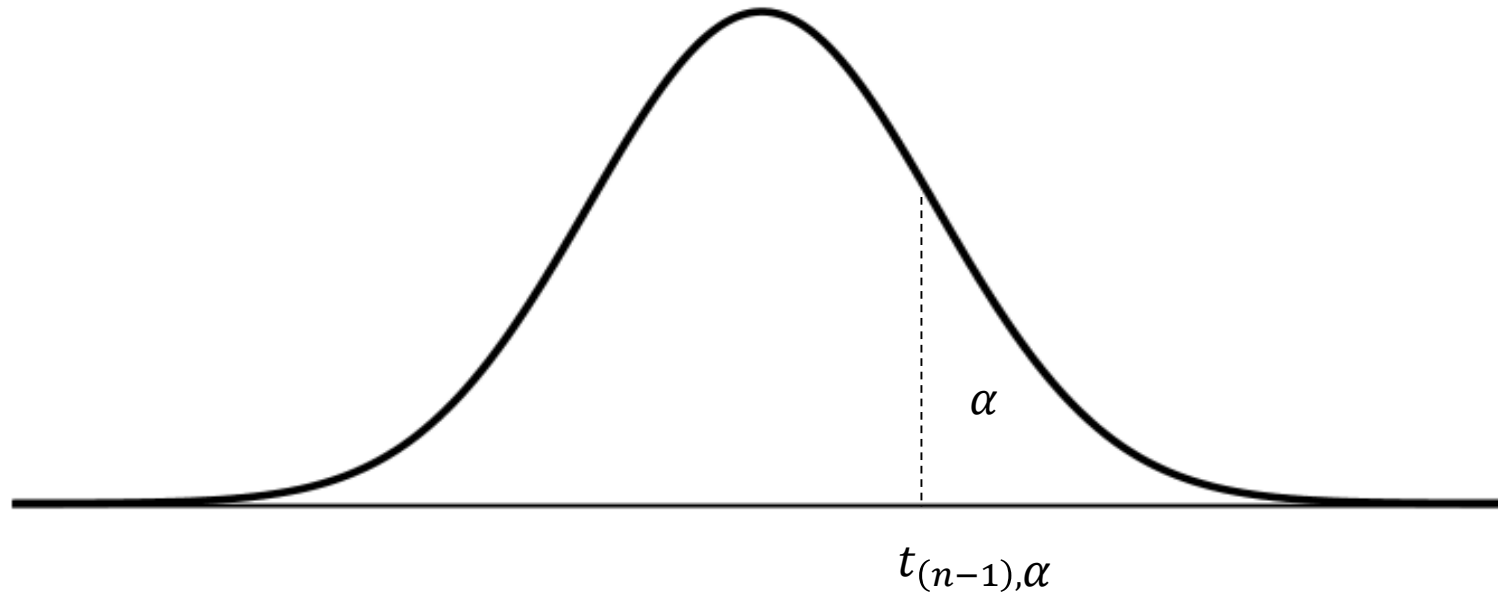
Since this is a two tail test,



If $t_{cal} > t_{(n-1), \frac{\alpha}{2}}$ or $t_{cal} < -t_{(n-1), \frac{\alpha}{2}}$ we reject H_0

Tests About Mean : t – test when population variance is unknown

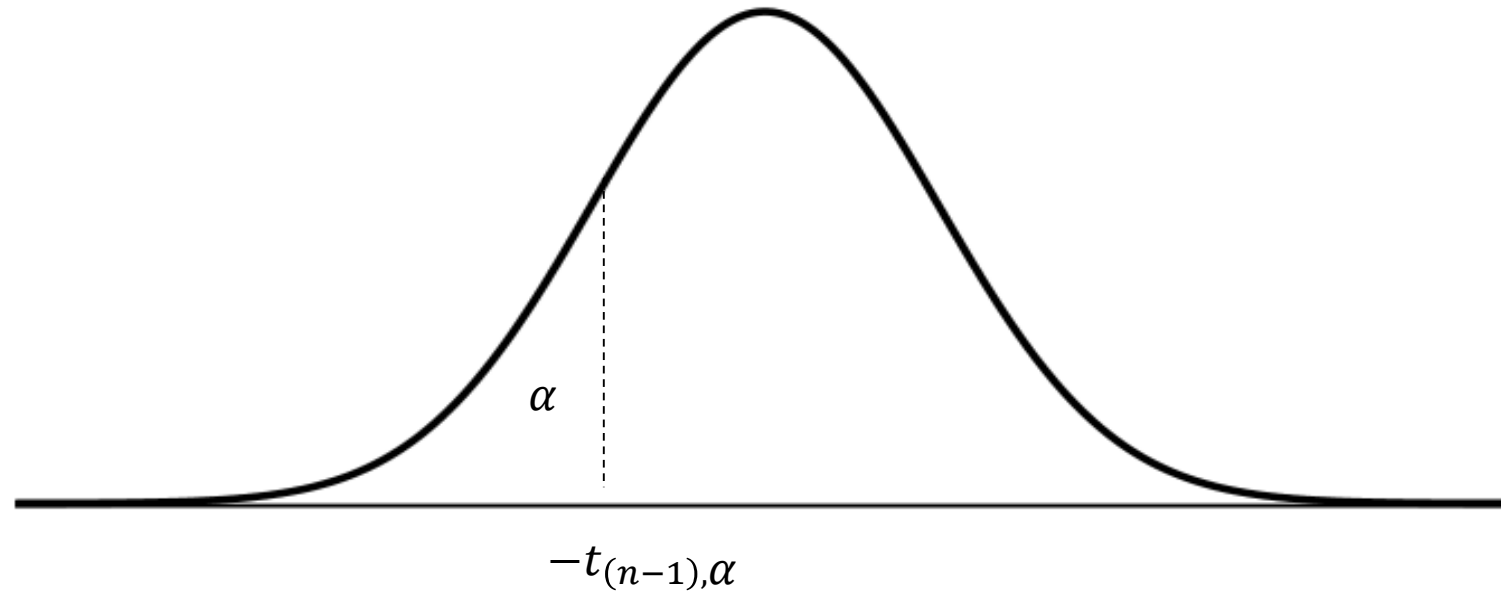
In a one tail – right tail test,, $H_0 : \mu = A$ and $H_1 : \mu > A$



If $t_{cal} > t_{(n-1),\alpha}$ we reject H_0

Tests About Mean : t – test when population variance is unknown

In a one tail – left tail test,, $H_0 : \mu = A$ and $H_1 : \mu < A$



If $t_{cal} < -t_{(n-1),\alpha}$ we reject H_0

Ex-

Five readings of the resistance in Ohms (Ω) of a piece of wire gave the following results.

1.15, 1.49, 1.54, 1.52, 1.54

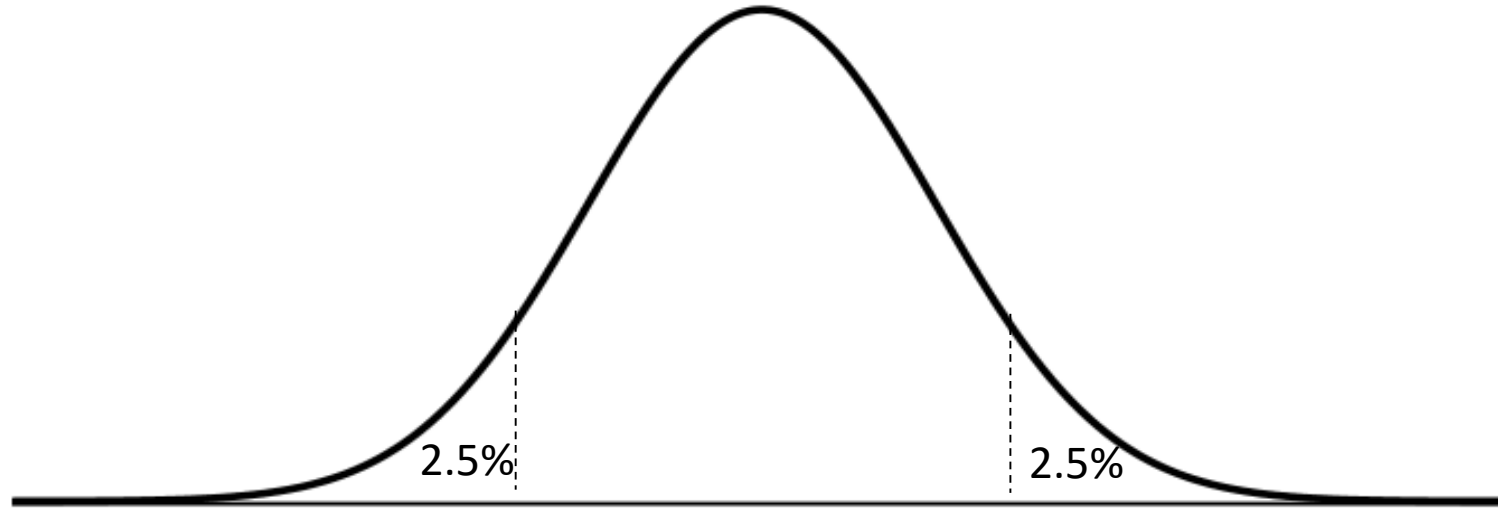
If the wire is pure silver, its resistance should be 1.5Ω . Test at 5% level of significance whether the wire is pure silver. Assume the normality of the readings.

$$H_0 : \mu = 1.5$$

$$H_1 : \mu \neq 1.5$$

Under H_0 ,

$$t_{cal} = \frac{(1.448 - 1.5)}{0.15/\sqrt{5}} = -0.78$$



$$-t_{(4),2.5\%} = -2.7763 \quad t_{(4),2.5\%} = 2.7763$$

$t_{cal} = -0.78 > -t_{(4),2.5\%}$, so we do not reject H_0 at 5% level of significance.

Conclusion - There is an evidence to say that the wire is pure silver at 5% level of significance.

cum. prob one-tail two-tails	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

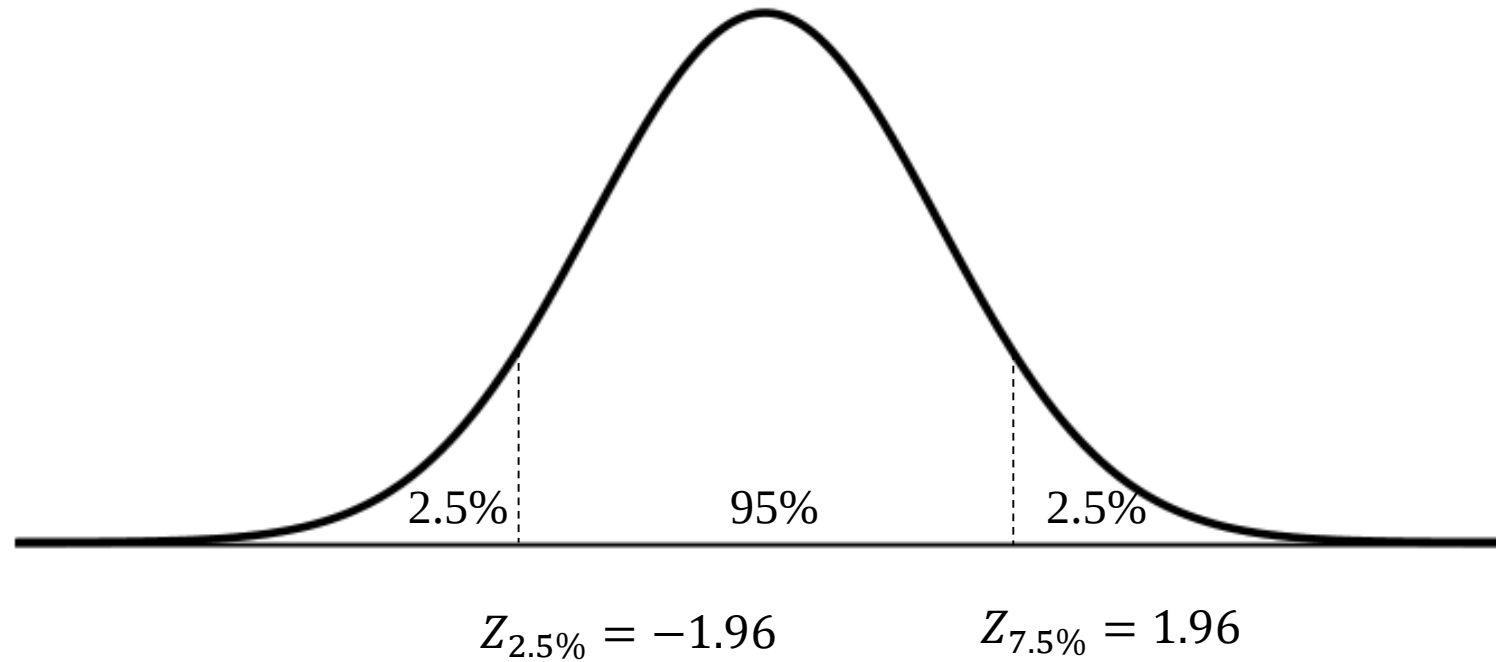
Confidence Interval Approach : Ex-

A random sample of 12 items is selected from a normal population with variance 4. Sample mean is 10.4. Test the following hypothesis at 5% level of significance.

$$H_0 : \mu = 15$$

$$H_1 : \mu \neq 15$$

Let's create the confidence interval for 5% level of significance under H_0 . 5% significance means 95% confidence should be there.



$$\left[10.4 - \frac{2}{\sqrt{12}} Z_{2.5\%}, 10.4 + \frac{2}{\sqrt{12}} Z_{2.5\%} \right] = [9.26, 11.52]$$

The parameter under H_0 lies outside the confidence interval. So H_0 will be rejected at 5% level of significance.

Conclusion - **There is an evidence to say that the population mean is not equal to 15 at 5% level of significance.**

P-Value Approach : Ex-

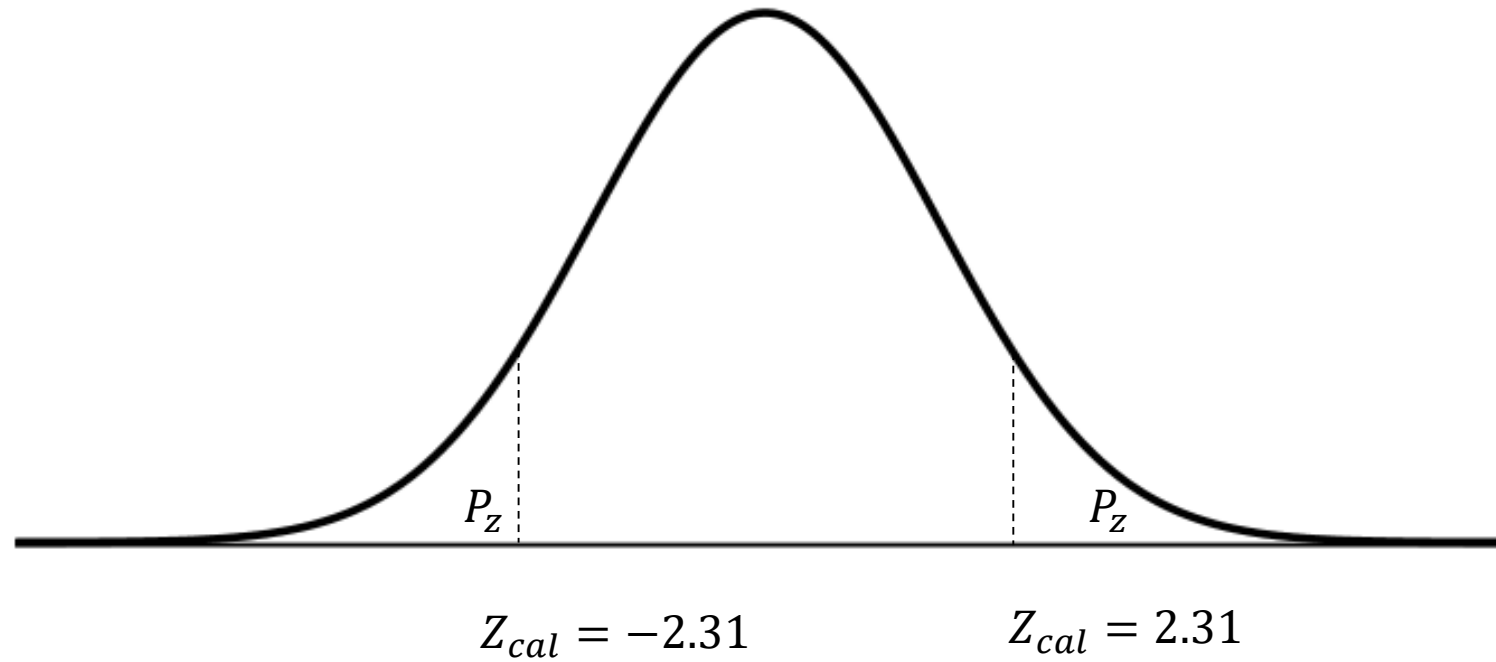
The average weight of all residents in town ABC is 168 lbs and the standard deviation is 3.9 lbs. A nutritionist believes the true mean to be different. She measured the weight of 36 individuals and found the mean to be 169.5. Test whether her idea is true or false at 5% level of significance.

$$H_0 : \mu = 168$$

$$H_1 : \mu \neq 168$$

Compute the test statistic under H_0 .

$$Z_{cal} = \frac{(169.5 - 168)}{3.9 / \sqrt{36}} = 2.31$$



$$\begin{aligned}P_z &= 0.01044 \\ \mathbf{P - Value} &= 2 \times P_z = 0.02088 \\ \mathbf{P - Value} &< \text{Significance Level} = 0.05\end{aligned}$$

So H_0 will be rejected at 5% level of significance.

Conclusion - **There is an evidence to say that the nutritionist's idea is true at 5% level of significance.**



Two Population Means



H.M. Samadhi Chathuranga Rathnayake

Properties of Expected Values

Consider X and Y are random variables follows a probability distribution.

$$E(X + Y) = E(X) + E(Y)$$

$$E(X - Y) = E(X) - E(Y)$$

$$E(kX) = kE(X)$$

Where k is a constant value.

$$E(aX \pm bY) = aE(X) \pm bE(Y)$$

Properties of Variances

Consider X and Y are random variables follows a probability distribution and they are **independent**.

$$V(X + Y) = V(X) + V(Y)$$

$$V(kX) = k^2V(X)$$

$$V(X - Y) = V(X + (-Y)) = V(X) + V(-Y) = V(X) + (-1)^2V(Y) = V(X) + V(Y)$$

Where k is a constant value.

$$V(aX \pm bY) = a^2V(X) + b^2V(Y)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case A - $\sigma_1^2 \neq \sigma_2^2$ but known

Let Y_1 and Y_2 be independent random variables distributed as $N(a, b)$ and $N(c, d)$ respectively. Then,
 $Y_1 \pm Y_2 \sim N(a \pm c, b + d)$.

Using the above property,

Let $x_1, x_2, x_3, \dots, x_n$ is a random sample from $N(\mu_1, \sigma_1^2)$ and $y_1, y_2, y_3, \dots, y_m$ is a random sample from $N(\mu_2, \sigma_2^2)$. We know that,

$$\bar{X} \sim N\left(\mu_1, \sigma_1^2/n\right) \quad \bar{Y} \sim N\left(\mu_2, \sigma_2^2/m\right)$$

So,

$$(\bar{X} - \bar{Y}) \sim N\left(\mu_1 - \mu_2, \sigma_1^2/n + \sigma_2^2/m\right)$$

So,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Then we can construct $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ as follows,

$$P\left(-Z_{\frac{\alpha}{2}} \leq Z \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-Z_{\frac{\alpha}{2}} \leq \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

So the $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} \sqrt{\sigma_1^2/n + \sigma_2^2/m}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case B - $\sigma_1^2 = \sigma_2^2$ but known

Let $x_1, x_2, x_3, \dots, x_n$ is a random sample from $N(\mu_1, \sigma^2)$ and $y_1, y_2, y_3, \dots, y_m$ is a random sample from $N(\mu_2, \sigma^2)$. We saw that for $\sigma_1^2 \neq \sigma_2^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

So the confidence interval can be derived as,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} \left(\sigma\sqrt{1/n + 1/m} \right)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case C - $\sigma_1^2 = \sigma_2^2$ but unknown

Let $x_1, x_2, x_3, \dots, x_n$ is a random sample from $N(\mu_1, \sigma^2)$ and $y_1, y_2, y_3, \dots, y_m$ is a random sample from $N(\mu_2, \sigma^2)$. We saw that for $\sigma_1^2 \neq \sigma_2^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{1/n + 1/m}} \sim N(0,1)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

We also know that,

$$\frac{(n-1)S_1^2}{\sigma^2} \sim \chi_{(n-1)}^2$$

$$\frac{(m-1)S_2^2}{\sigma^2} \sim \chi_{(m-1)}^2$$

Let Y_1 and Y_2 be independent random variables distributed as $\chi_{(a)}^2$ and $\chi_{(b)}^2$ respectively. Then, $Y_1 + Y_2 \sim \chi_{(a+b)}^2$.

Using the above property,

$$W = \frac{(n-1)S_1^2}{\sigma^2} + \frac{(m-1)S_2^2}{\sigma^2} \sim \chi_{(m+n-2)}^2$$

Because $\frac{(n-1)S_1^2}{\sigma^2}$ and $\frac{(m-1)S_2^2}{\sigma^2}$ are independent.

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Using the definition of t distribution,

$$T = \frac{Z}{\sqrt{\frac{W}{n+m-2}}} \sim t_{(n+m-2)}$$

$$T = \frac{\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{1/n + 1/m}}}{\sqrt{\frac{\frac{(n-1)S_1^2}{\sigma^2} + \frac{(m-1)S_2^2}{\sigma^2}}{n+m-2}}} = \frac{\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{1/n + 1/m}}}{\sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}} = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{1/n + 1/m}} \sim t_{(n+m-2)}$$

Then we can construct $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ as follows,

$$P \left(-t_{(n+m-2), \frac{\alpha}{2}} \leq T \leq t_{(n+m-2), \frac{\alpha}{2}} \right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

So the $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm t_{(n+m-2), \frac{\alpha}{2}} S_p \sqrt{1/n + 1/m}$$

If n and m are very large (say 30), the $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ can be written as,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} S_p \sqrt{1/n + 1/m}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case D - $\sigma_1^2 \neq \sigma_2^2$ but unknown

The $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm t_{(df), \frac{\alpha}{2}} \sqrt{S_1^2/n + S_2^2/m}$$

$$df = \frac{\left(S_1^2/n + S_2^2/m\right)^2}{\frac{\left(S_1^2/n\right)^2}{n-1} + \frac{\left(S_2^2/m\right)^2}{m-1}}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Ex-

From a large class 4 students were drawn and their Maths marks were 64, 66, 89 and 77. From a second large class 3 students were drawn and their Maths marks were 56, 71 and 53. Assuming the classes are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the two Maths marks.

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Ex-

From a large class 4 students were drawn and their Maths marks were 64, 66, 89 and 77. From a second large class 3 students were drawn and their Maths marks were 56, 71 and 53. Assuming the classes are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the two Maths marks.

$$\bar{X} = 74 \text{ and } \bar{Y} = 60$$

$$S_1^2 = 132.67 \text{ and } S_2^2 = 92$$

$$S_p = \sqrt{\frac{3 \times 132.67 + 2 \times 92}{4 + 3 - 2}} = 10.78$$

$$t_{2.5\%,5} = 2.57$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

$$(\bar{X} - \bar{Y}) \pm t_{(n+m-2), \frac{\alpha}{2}} S_p \sqrt{1/n + 1/m}$$

$$14 \pm 10.78 \sqrt{\frac{1}{4} + \frac{1}{3}} \times 2.57$$

$$[-7.034, 35.09]$$

Comparing two population means.

Consider following hypotheses,

$$H_0 : \mu_1 - \mu_2 = k , H_1 : \mu_1 - \mu_2 \neq k$$

$$H_0 : \mu_1 - \mu_2 = k , H_1 : \mu_1 - \mu_2 > k$$

$$H_0 : \mu_1 - \mu_2 = k , H_1 : \mu_1 - \mu_2 < k$$

Consider $\mu_{def} = \mu_1 - \mu_2$,

$$H_0 : \mu_{def} = k , H_1 : \mu_{def} \neq k$$

$$H_0 : \mu_{def} = k , H_1 : \mu_{def} > k$$

$$H_0 : \mu_{def} = k , H_1 : \mu_{def} < k$$

This is pretty much similar to one population case. Only difference can be seen in the test statistic.

When we compare two population means we have to check where the null hypothesis is $\mu_1 = \mu_2$ situation which is $\mu_1 - \mu_2 = \mu_{def} = 0$ (k=0)

Comparing two population means – Unpaired case.

Let $x_1, x_2, x_3, \dots, x_n$ is a random sample from $N(\mu_1, \sigma_1^2)$ and $y_1, y_2, y_3, \dots, y_m$ is a random sample from $N(\mu_2, \sigma_2^2)$. We know that,

$$\bar{X} \sim N\left(\mu_1, \sigma_1^2/n\right)$$

$$\bar{Y} \sim N\left(\mu_2, \sigma_2^2/m\right)$$

So,

$$(\bar{X} - \bar{Y}) \sim N\left(\mu_1 - \mu_2, \sigma_1^2/n + \sigma_2^2/m\right)$$

This is the sampling distribution of difference between two sample means. Then we can use **z** or **t** for known or unknown population variances respectively as the test statistic. Following sub cases explain the test statistic for different conditions of population variances.

Comparing two population means – Unpaired case.

Case A - $\sigma_1^2 \neq \sigma_2^2$ but known

Then the test statistic is,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

Case B - $\sigma_1^2 = \sigma_2^2$ but known

Then the test statistic is,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{1/n + 1/m}} \sim N(0,1)$$

Comparing two population means – Unpaired case.

Case C - $\sigma_1^2 = \sigma_2^2$ but unknown

Then the test statistic is,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{1/n + 1/m}} \sim t_{(n+m-2)}$$

Where,

$$S_p = \sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$$

Comparing two population means – Unpaired case.

Case D - $\sigma_1^2 \neq \sigma_2^2$ but unknown

Then the test statistic is,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n + S_2^2/m}} \sim t_{(df)}$$

Where,

$$df = \frac{\left(S_1^2/n + S_2^2/m\right)^2}{\frac{\left(S_1^2/n\right)^2}{n-1} + \frac{\left(S_2^2/m\right)^2}{m-1}}$$

Comparing two population means – Unpaired case.

Ex-

A test was given to a group of 100 scouts and 144 guides. The mean score for scouts was 27.53 and the mean score for the guides was 26.81. Assuming a common population standard deviation of 3.48, test at 5% level of significance whether the scouts performance in the test was better than that of the guides. Assume that the scores are normally distributed.

Comparing two population means – Unpaired case.

Ex-

A test was given to a group of 100 scouts and 144 guides. The mean score for scouts was 27.53 and the mean score for the guides was 26.81. Assuming a common population standard deviation of 3.48, test at 5% level of significance whether the scouts performance in the test was better than that of the guides. Assume that the scores are normally distributed.

$$n_1 = 100 \quad n_2 = 144$$

$$\bar{X} = 27.53 \quad \bar{Y} = 26.81$$

$$\sigma = 3.48$$

$$H_0 : \mu_1 - \mu_2 = 0, H_1 : \mu_1 - \mu_2 > 0$$

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{1/n + 1/m}} \sim N(0,1)$$

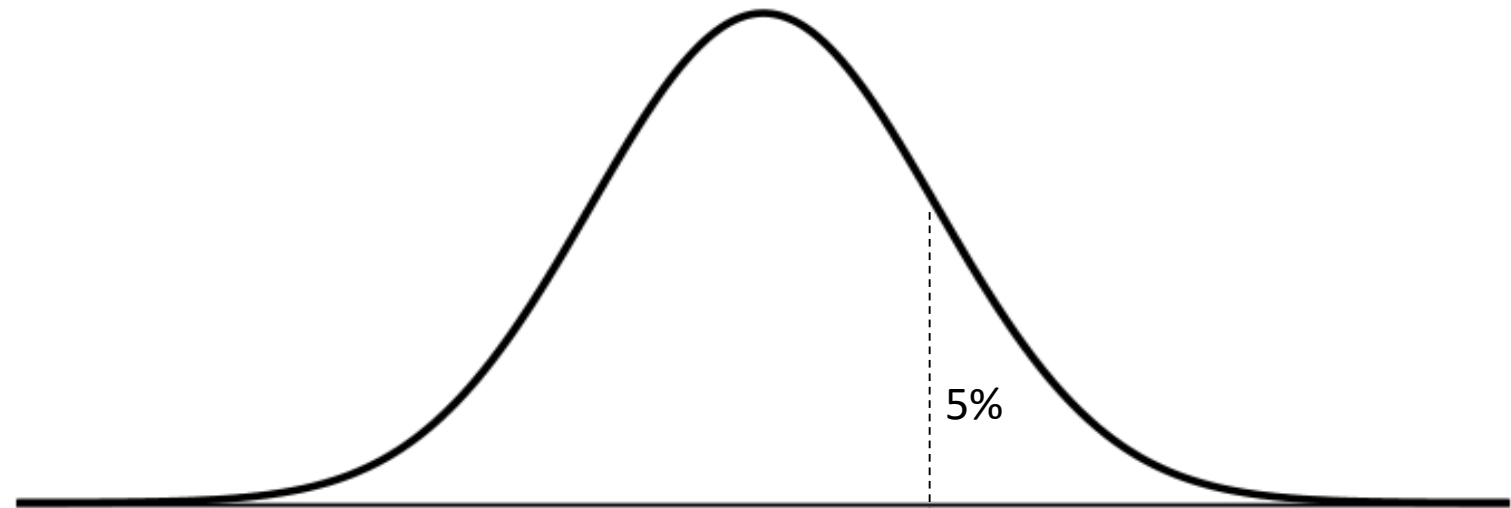
Comparing two population means – Unpaired case.

Under H_0 ,

$$Z_{cal} = \frac{(27.53 - 26.81) - (0)}{3.41 \sqrt{1/100 + 1/144}} = 1.5894$$

This is one tail right tail test.

$$Z_{cal} < Z_{critical}$$



$$Z_{critical} = 1.645$$

Comparing two population means – Unpaired case.

Do not reject H_0 5% level of significance.

Conclusion - There is no evidence to say that the scouts performance in the test was better than that of the guides at 5% level of significance.

Comparing two population means – Paired case.

The objective of this section is testing the difference between population means based on means of two dependent samples.

Paired samples: The sample selected from the first population is related to the corresponding sample from the second population.

Assumptions we made under paired samples:

The difference of the pairs follow a normal distribution or the number of pairs is large. (note here that if the number of pairs is less than 30, we need to check whether the **differences** are normally distributed, but we do not need to check for the normality of each population.)

The key to analyze a matched sample design is to use the difference data.

Comparing two population means – Paired case.

Consider hypothesis,

$$H_0 : \mu_{def} = k , H_1 : \mu_{def} \neq k$$

$$H_0 : \mu_{def} = k , H_1 : \mu_{def} > k$$

$$H_0 : \mu_{def} = k , H_1 : \mu_{def} < k$$

Test statistic:

Let d =differences between the pairs of data, then $\bar{d}$ =mean of these differences. Then the test statistic is,

$$T = \frac{\bar{d} - \mu_{def}}{S_{def} / \sqrt{n}} \sim t_{(n-1)}$$

Where n denotes the number of pairs or the number of differences.

Comparing two population means – Paired case.

Ex-

Trace metals in drinking water affect the flavor and unusually high concentration can pose a health hazard. Ten pairs of data were taken measuring zinc concentration in bottom water and surface water. Does the data suggest that the true average concentration in the bottom water exceeds that of surface water? (5% level of significance)

	Location									
	1	2	3	4	5	6	7	8	9	10
Zinc concentration in bottom water	0.430	0.266	0.567	0.531	0.707	0.716	0.651	0.589	0.469	0.723
Zinc concentration in surface water	0.415	0.238	0.390	0.410	0.605	0.609	0.632	0.523	0.411	0.612

Comparing two population means – Paired case.

Since the sample size is not large enough (Less than 30), we need to check whether the differences follow a normal distribution.

	B	S	D
0	0.430	0.415	0.015
1	0.266	0.238	0.028
2	0.567	0.390	0.177
3	0.531	0.410	0.121
4	0.707	0.605	0.102
5	0.716	0.609	0.107
6	0.651	0.632	0.019
7	0.589	0.523	0.066
8	0.469	0.411	0.058
9	0.723	0.612	0.111

Thus, we can test and show that the difference may come from a normal distribution

Comparing two population means – Paired case.

$$H_0 : \mu_{def} = 0 , H_1 : \mu_{def} > 0$$

$$T_{cal} = 4.86$$

$$t_{0.05,9} = 1.833$$

Since $T_{cal} > t_{0.05,9}$, Reject H_0 at 5% level of significance.

Conclusion - There is an evidence to say that the true average concentration in the bottom water exceeds that of surface water at 5% level of significance.



Analysis Of Variance (ANOVA)



H.M. Samadhi Chathuranga Rathnayake

Analysis of Variance (ANOVA)

An extension of the pooled variance t-test for the difference between two means. Though the term mentions the word variance, we are testing differences in means. ANOVA separates total variation into variation within (random variation) and among the groups. So long as the ratio between within and among variation is large enough, we conclude that there are significant differences between groups.

Assumptions:

- Populations are normally distributed.
- Populations have equal variances.
- Samples are randomly and independently drawn.

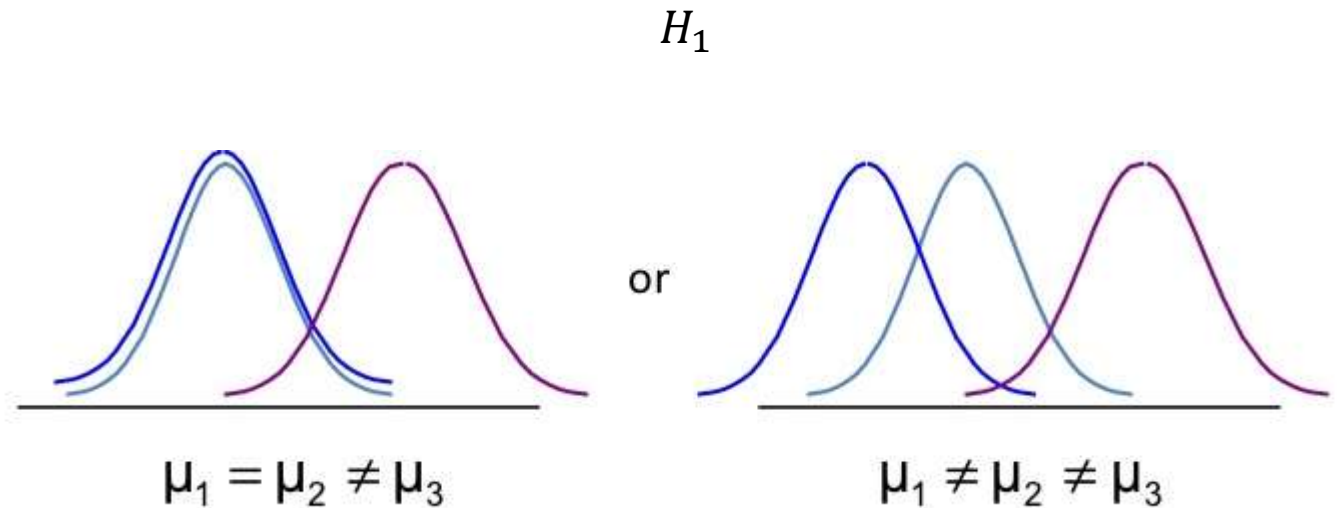
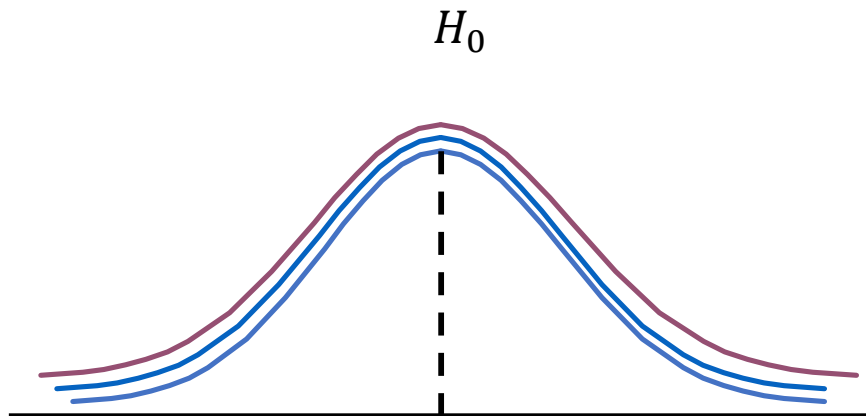
One Way ANOVA

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_c$$

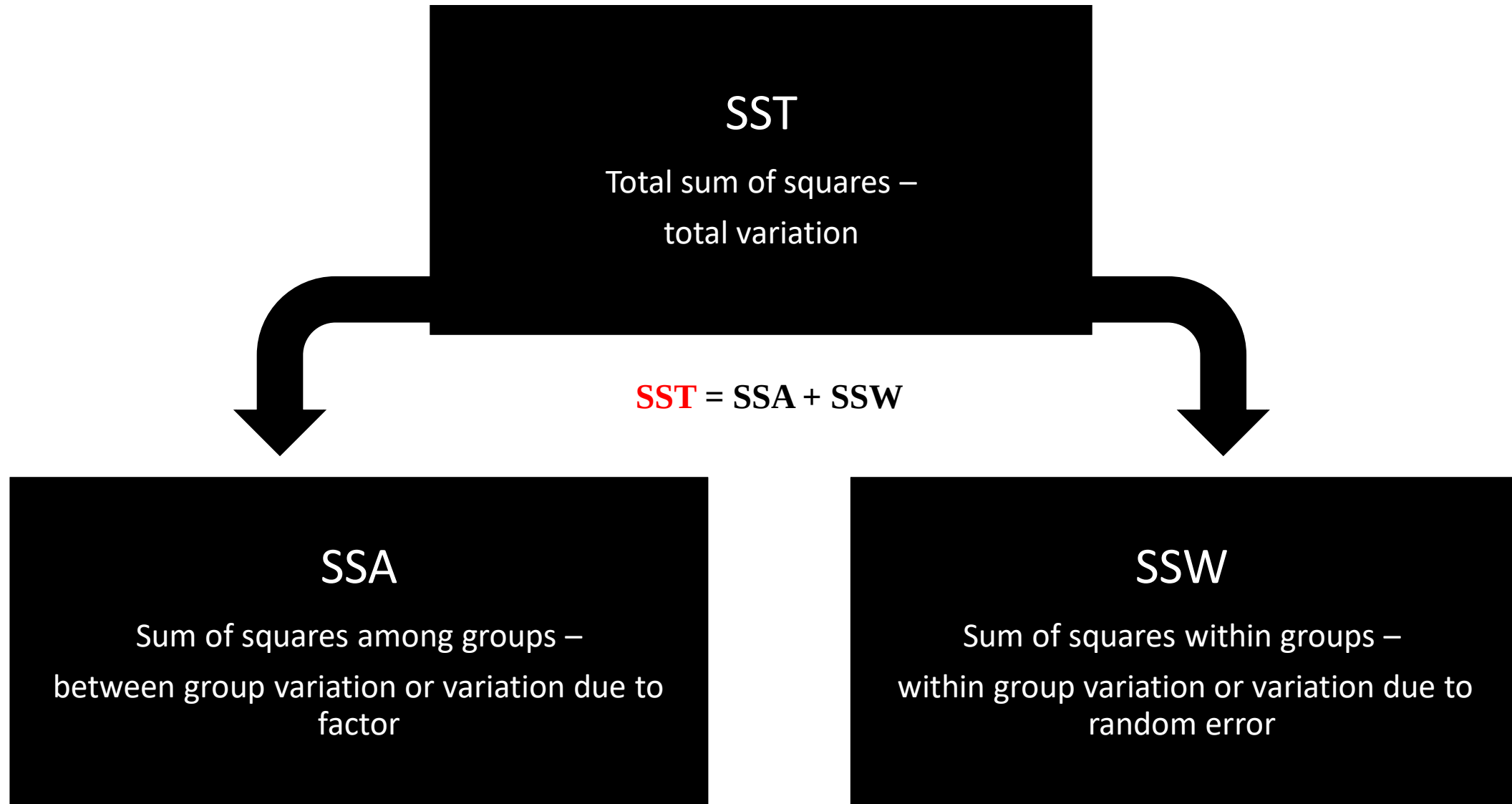
All population means are equal. (no variation in means among groups)

H_1 : At least one population mean is different

At least one population mean is different. (not necessarily that all population means are different)



One Way ANOVA



One Way ANOVA

$$SST = \sum_{j=1}^c \sum_{i=1}^{n_j} (X_{ij} - \bar{\bar{X}})^2$$

- c = number of groups or levels
- n_j = number of observations in group j
- X_{ij} = i^{th} observation from group j
- $\bar{\bar{X}}$ = grand mean (mean of all data values)

One Way ANOVA

$$SSA = \sum_{j=1}^c n_j (\bar{X}_j - \bar{\bar{X}})^2$$

- $\bar{X}_j$ = sample mean from group j

$$\text{Mean Square Among (MSA)} = \frac{SSA}{c - 1}$$

- where c-1 is the degrees of freedom of the SSA

One Way ANOVA

$$SSW = \sum_{j=1}^c \sum_{i=1}^{n_j} (X_{ij} - \bar{X}_j)^2$$

- $\bar{X}_j$ = sample mean from group j

$$\text{Mean Square Within (MSW)} = \frac{SSW}{n - c}$$

- Where n-c is the degrees of freedom of the SSW

One Way ANOVA Table

Source of variation	Degrees of freedom	Mean square	F
Between groups	c-1	$MSA = \frac{SSA}{c - 1}$	$F_{stat} = \frac{MSA}{MSW}$
Within groups	n-c	$MSW = \frac{SSW}{n - c}$	
Total	n-c+(c-1)=n-1	$MST = \frac{SST}{n - 1}$	

One Way ANOVA Test

The test statistic is the ratio of the mean of among variance and mean of within variance. The ratio must always be positive.

- $c-1$ is typically small
- $n-c$ is typically large

Decision rule: Reject H_0 if $F_{stat} > F_{\alpha, c-1, n-c}$. Otherwise, do not reject.

One Way ANOVA Test

Ex:-

You want to see if three different tennis rackets hit the ball different distances. You randomly select five measurements from trials on an automated ball machine. At the 5% significance level, is there a difference in mean distance?

Racket 1	Racket 2	Racket 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

One Way ANOVA Test

Ex:-

When the needful calculations are done,

$$\bar{\bar{X}} = 227 \quad \bar{X}_1 = 249.2 \quad \bar{X}_2 = 226 \quad \bar{X}_3 = 205.8 \quad n_1 = n_2 = n_3 = 5 \quad n = 15 \quad c = 3$$

$$SSA = 4,716.4$$

$$SSW = 1,119.6$$

$$MSA = 4716.4/(3 - 1) = 2,358.2$$

$$MSW = 1119.6/(15 - 3) = 93.3$$

$$F_{stat} = \frac{2358.2}{93.3} = 25.275, \quad F_{0.05,2,12} = 3.89$$

Since $F_{stat} > F_{0.05,2,12}$, Reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say that at least one mean is different from the rest at 5% level of significance.

Checking Homogeneity of Variance

Levene's test is used for testing the assumption that the variances of each population are equal (assumption of ANOVA).

The test involves three steps:

1. Define the null and alternative hypothesis.

$$H_0: \sigma_1^2 = \sigma_2^2 = \dots = \sigma_c^2 \quad \& \quad H_1: \text{Not all } \sigma_j^2 \text{ are equal (At least one variance is different)}$$

2. Compute the absolute value of the difference between each value and the median of each group.
3. Perform one-way ANOVA on these absolute differences.

Checking Homogeneity of Variance

Ex:-

Consider the above discussed example.

$$H_0: \sigma_1^2 = \sigma_2^2 = \sigma_3^2 \quad \& \quad H_1: \text{Not all } \sigma_j^2 \text{ are equal}$$

Racket 1	Racket 2	Racket 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

Colored cells are the median values.

Absolute difference from median		
Racket 1	Racket 2	Racket 3
3	7	4
12	9	18
10	8	7
14	0	2
0	11	0

Checking Homogeneity of Variance

Ex:-

$$F_{stat} = 0.92 \quad , \quad F_{0.05,2,12} = 3.89$$

Since $F_{stat} < F_{0.05,2,12}$, Do not reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say the homogeneity of variance assumption is satisfied at 5% level of significance.

Post – Hoc Analysis : Mean Comparisons

When the null hypothesis of ANOVA is rejected, it only suggests that there are significant differences between means. However, it does not specify which of the treatment means differ. Therefore further comparisons between group or treatment is required.



Post – Hoc Analysis : The Fisher Least Significant Difference (Fisher's LSD)

This is the most commonly used tool for testing the significance between all possible pairs of means. This method ensures error rate (Significance Level α) for each individual pair wise comparison.

The method is to declare the difference between any pair of means significant at $\alpha\%$ level if the,

$$|Observed\ Difference| > LSD$$

Where,

$$LSD = t_{df, \frac{\alpha}{2}} \sqrt{MSW \left(\frac{1}{n_j} + \frac{1}{n_i} \right)}$$

Where df=Error Degrees of Freedom

Post – Hoc Analysis : The Fisher Least Significant Difference (Fisher’s LSD)

Ex:-

Doughnuts are supposed to absorb fat in various amounts, during cooking. In an experiment to compare 4 types of fat, 6 batches of Doughnuts were prepared from each type. The measurement was the amount (g) of fat absorbed. The data were analyzed as for a one-way treatment structure in a CRD. The following is the summary of data.

Treatment	Fat 01	Fat 02	Fat 03	Fat 04
r	6	6	6	6
mean	72	85	76	62

Post – Hoc Analysis : The Fisher Least Significant Difference (Fisher's LSD)

Ex:-

$$MSW = 100.9 \quad df = 20$$

So,

$$LSD = t_{20, \frac{0.05}{2}} \sqrt{100.9 \left(\frac{1}{6} + \frac{1}{6} \right)} = 2.086 \times 5.80 = 12.1$$

Any pair with mean difference greater than 12.1 will be declared significant at 5% level of significance.

$$\bar{X}_{Fat1} = 72$$

$$\bar{X}_{Fat2} = 85$$

$$\bar{X}_{Fat3} = 76$$

$$\bar{X}_{Fat4} = 62$$

Ex:

$$|\bar{X}_2 - \bar{X}_1| = 85 - 72 = 13 > LSD$$

So the difference of the means of Fat 01 & Fat 02 is significant at 5% level of significance. We can check this for other pairs too.



Population Proportions



H.M. Samadhi Chathuranga Rathnayake

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

Consider a population in which the proportion of success is p . If a random sample of n is taken from this population and then number of successes in the sample (X) allows the binomial distribution.

$$X \sim \text{Bin}(n, p)$$

Now for large n ,

$$X \sim N(np, npq) \quad \text{where } q = 1 - p$$

Now consider $\hat{P} = \frac{X}{n}$. Then,

$$E(\hat{P}) = p \quad \text{and} \quad V(\hat{P}) = \frac{pq}{n}$$

So, for large samples ($n > 30$),

$$\hat{P} \sim N\left(p, \frac{pq}{n}\right)$$

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

By standardizing this statistic we get that,

$$Z = \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1)$$

Then we can construct a $(1-\alpha)100\%$ Confidence Interval for p as follows.

$$P\left(-Z_{\frac{\alpha}{2}} \leq Z \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-Z_{\frac{\alpha}{2}} \leq \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

So the confidence interval will be,

$$\left[\hat{P} - Z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}} , \quad \hat{P} + Z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}} \right] \approx \left[\hat{P} - Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} , \quad \hat{P} + Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \right]$$

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

Ex-

A manufacturer wants to assess the proportion of defective items in a large batch produced by a particular machine. A random sample of 300 items is taken and it is observed that 45 items of them are defective. Calculate a 95% confidence interval for the proportion of defective items in the complete batch.

$(1-\alpha)100\%$ Confidence Interval for population proportion of success for large samples

Ex-

A manufacturer wants to assess the proportion of defective items in a large batch produced by a particular machine. A random sample of 300 items is taken and it is observed that 45 items of them are defective. Calculate a 95% confidence interval for the proportion of defective items in the complete batch.

$$\hat{p} = \frac{45}{300} = 0.15$$

$$\hat{q} = 0.85$$

$$\hat{P} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} = 0.15 \pm 1.96 \sqrt{\frac{0.15 \times 0.85}{300}}$$

$$[0.11, 0.19]$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Consider we have taken two samples from two populations. Then for large sample sizes,

$$\hat{p}_1 \sim N\left(p_1, \frac{p_1 q_1}{n_1}\right)$$

$$\hat{p}_2 \sim N\left(p_2, \frac{p_2 q_2}{n_2}\right)$$

Then,

$$(\hat{p}_1 - \hat{p}_2) \sim N\left(p_1 - p_2, \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}\right)$$

So,

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}} \sim N(0,1)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Then we can construct a $(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success as follows.

$$P\left(-Z_{\frac{\alpha}{2}} \leq Z \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-Z_{\frac{\alpha}{2}} \leq \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}} \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

So the confidence interval will be,

$$\left[(\hat{p}_1 - \hat{p}_2) - Z_{\frac{\alpha}{2}} \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}, (\hat{p}_1 - \hat{p}_2) + Z_{\frac{\alpha}{2}} \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}} \right] \approx$$
$$\left[(\hat{p}_1 - \hat{p}_2) - Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}, (\hat{p}_1 - \hat{p}_2) + Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \right]$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Ex:-

A clinic trial is conducted to determine if a certain type of inoculation has an effect on the incident of a certain disease. A sample of 1000 rats was kept in a controlled environment for a period of 1 year and 500 rats were given the inoculation. Of the group not given the drug, there was 120 incidences of the disease, while 98 of the inoculated group contracted it. Compute the 90% confidence interval for the difference between the probability of incidents of the disease in uninoculated rats and the probability of incidents after receiving the drug.

$(1-\alpha)100\%$ Confidence Interval for the difference between two population proportions of success for large samples

Ex:-

A clinic trial is conducted to determine if a certain type of inoculation has an effect on the incident of a certain disease. A sample of 1000 rats was kept in a controlled environment for a period of 1 year and 500 rats were given the inoculation. Of the group not given the drug, there was 120 incidences of the disease, while 98 of the inoculated group contracted it. Compute the 90% confidence interval for the difference between the probability of incidents of the disease in uninoculated rats and the probability of incidents after receiving the drug.

$$\hat{p}_1 = \frac{120}{500} = 0.24 \quad \hat{p}_2 = \frac{98}{500} = 0.196$$

$$(\hat{p}_1 - \hat{p}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} = 0.044 \pm 1.65 \times \sqrt{\frac{0.24 \times 0.76}{500} + \frac{0.196 \times 0.804}{500}} = 0.044 \pm 0.043$$

[0.001, 0.087]

Test involving one sample proportion

Consider following hypotheses,

$$H_0 : p = k , H_1 : p \neq k$$

$$H_0 : p = k , H_1 : p > k$$

$$H_0 : p = k , H_1 : p < k$$

Where p represents the population proportion of success and $0 \leq k \leq 1$. Then the test statistic is,

$$Z = \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} \sim N(0,1)$$

Test involving one sample proportion

Ex:-

From planted 100 seeds, 83 seeds were germinated . Test the manufacturers claim, the seeds are highly productive with 90% germination ratio. Use 5% level of significance.

Test involving one sample proportion

Ex:-

From planted 100 seeds, 83 seeds were germinated . Test the manufacturers claim, the seeds are highly productive with 90% germination ratio. Use 5% level of significance.

$$H_0 : p = 0.9 , H_1 : p \neq 0.9$$

$$Z_{cal} = \frac{\hat{P} - p}{\sqrt{\frac{pq}{n}}} = \frac{0.83 - 0.9}{\sqrt{\frac{0.9 \times 0.1}{100}}} = -2.33$$

$$Z_{2.5\%} = 1.96 \text{ and } -Z_{2.5\%} = -1.96$$

$$Z_{cal} < -1.96$$

So reject H_0 at 5% level of significance.

Conclusion: There is an evidence to say that the claim of the manufacturer is not true at 5% level of significance.

Test involving two population proportions

Consider following hypotheses,

$$H_0 : p_1 - p_2 = k , H_1 : p_1 - p_2 \neq k$$

$$H_0 : p_1 - p_2 = k , H_1 : p_1 - p_2 > k$$

$$H_0 : p_1 - p_2 = k , H_1 : p_1 - p_2 < k$$

Where p represents the population proportion of success and $0 \leq k \leq 1$.

We know that,

$$(\hat{p}_1 - \hat{p}_2) \sim N \left(p_1 - p_2, \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2} \right)$$

There are two different test statistics for this parameter, the choice of which are to be used depends on the value of **k**.

Test involving two population proportions

Case 01: $k = 0$

The test statistic is,

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1)$$

$$\hat{p}_1 = \frac{x_1}{n_1} \text{ (The proportion of success in sample 01)}$$

$$\hat{p}_2 = \frac{x_2}{n_2} \text{ (The proportion of success in sample 02)}$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2} \text{ (The common proportion of success in sample 01 \& 02) and } \hat{q} = 1 - \hat{p}$$

Test involving two population proportions

Case 02: $k \neq 0$

The test statistic is,

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}} \sim N(0,1)$$

Test involving two population proportions

Ex:-

In a manufacturing process, two different methods A and B were employed. To evaluate both of manufacturing processes, the manager decided to examine the differences between them. He took 3740 items from each process and 60 defectives were found in process A and 28 in process B. Test at 5% level whether process B is better than process A.

Test involving two population proportions

Ex:-

In a manufacturing process, two different methods A and B were employed. To evaluate both of manufacturing processes, the manager decided to examine the differences between them. He took 3740 items from each process and 60 defectives were found in process A and 28 in process B. Test at 5% level whether process B is better than process A.

$$H_0 : p_B - p_A = 0, H_1 : p_B - p_A < 0$$

$$\hat{p}_A = \frac{60}{3740} = 0.02 \quad \hat{p}_B = \frac{28}{3740} = 0.0075 \quad \hat{p} = \frac{60 + 28}{7480} = 0.01 \quad \hat{q} = 0.99$$

$$Z = \frac{(\hat{p}_B - \hat{p}_A) - (p_B - p_A)}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_B} + \frac{1}{n_A}\right)}} = \frac{(0.0075 - 0.02) - 0}{\sqrt{0.01 \times 0.99 \times \left(\frac{1}{3740} + \frac{1}{3740}\right)}} = -5.43$$

$$-5.43 < -Z_{5\%} = -1.65$$

So reject H_0 at 5% level of significance.

Conclusion : There is an evidence to say that the process B is better than the process A at 5% level of significance



Population Variances



H.M. Samadhi Chathuranga Rathnayake

$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$. We know that for a sample of a population,

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_{(n)}^2$$

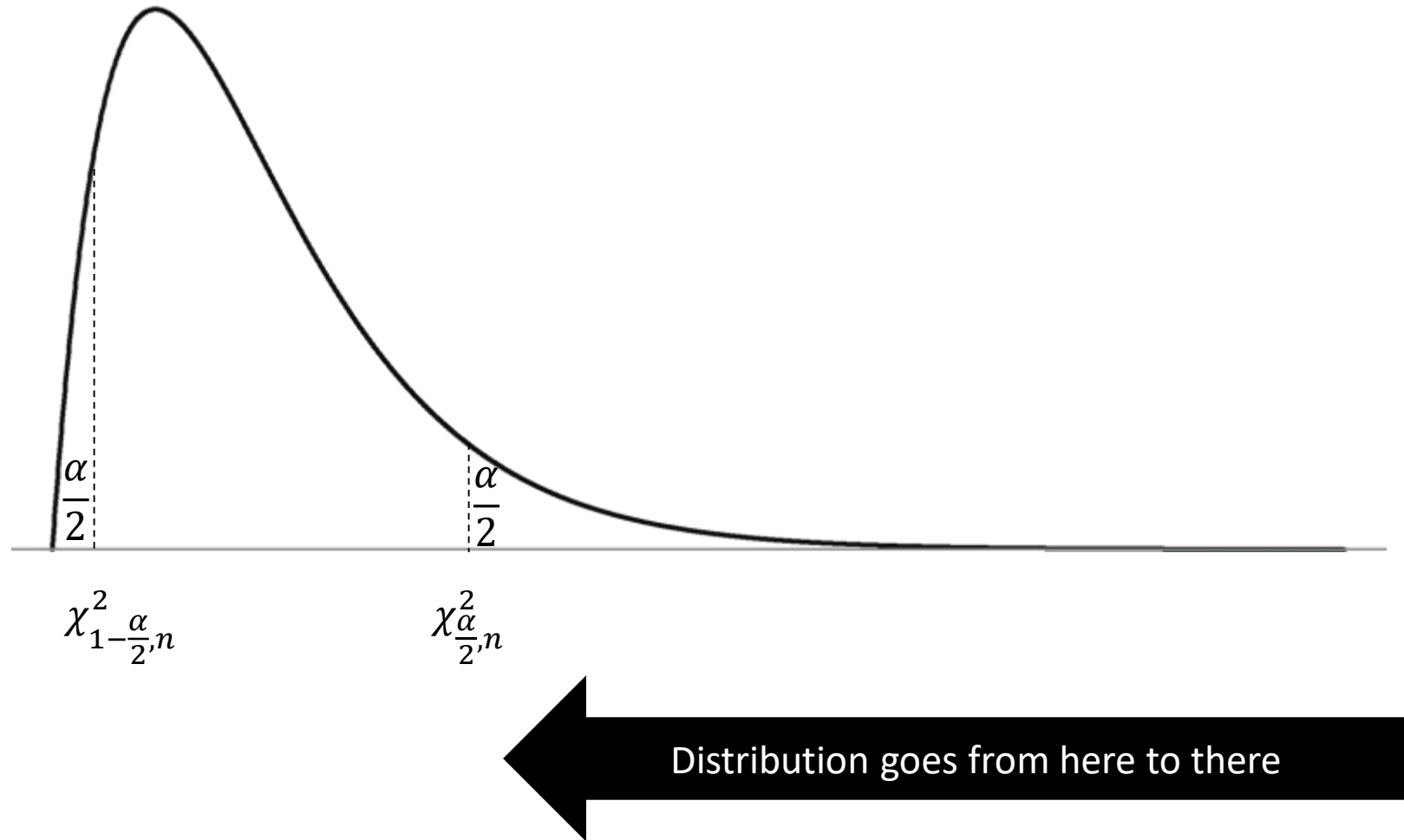
Consider this as the pivotal quantity. Then,

$$P \left(L \leq \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \leq U \right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

In the Chi-Squared distribution,



$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

Then,

$$P\left(\chi_{1-\frac{\alpha}{2},n}^2 \leq \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \leq \chi_{\frac{\alpha}{2},n}^2\right) = 1 - \alpha$$

$$P\left(\frac{1}{\chi_{1-\frac{\alpha}{2},n}^2} \geq \frac{\sigma^2}{\sum_{i=1}^n (X_i - \mu)^2} \geq \frac{1}{\chi_{\frac{\alpha}{2},n}^2}\right) = 1 - \alpha$$

$$P\left(\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{1-\frac{\alpha}{2},n}^2} \geq \sigma^2 \geq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{\frac{\alpha}{2},n}^2}\right) = 1 - \alpha$$

$$P\left(\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{\frac{\alpha}{2},n}^2} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{1-\frac{\alpha}{2},n}^2}\right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

Then $(1-\alpha)100\%$ confidence interval for variance when the population mean is known is,

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{\frac{\alpha}{2}, n}^2}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{1-\frac{\alpha}{2}, n}^2} \right]$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

We know that for a sample of a population,

$$\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

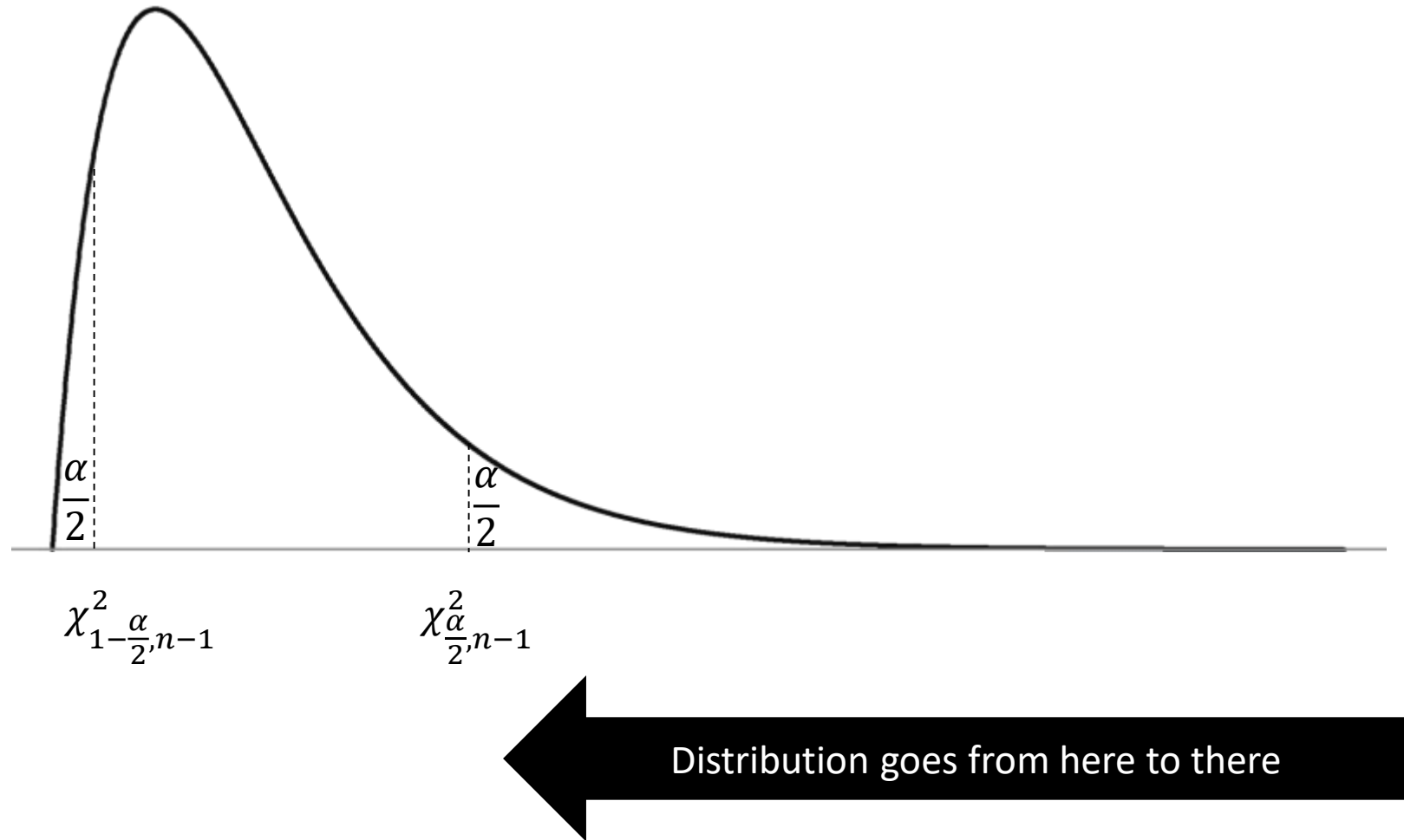
Consider this as the pivotal quantity. Then,

$$P \left(L \leq \frac{(n-1)S^2}{\sigma^2} \leq U \right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

In the Chi-Squared distribution,



$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

Then,

$$P\left(\chi_{1-\frac{\alpha}{2},n-1}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{\frac{\alpha}{2},n-1}^2\right) = 1 - \alpha$$

$$P\left(\frac{1}{\chi_{1-\frac{\alpha}{2},n-1}^2} \geq \frac{\sigma^2}{(n-1)S^2} \geq \frac{1}{\chi_{\frac{\alpha}{2},n-1}^2}\right) = 1 - \alpha$$

$$P\left(\frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2},n-1}^2} \geq \sigma^2 \geq \frac{(n-1)S^2}{\chi_{\frac{\alpha}{2},n-1}^2}\right) = 1 - \alpha$$

$$P\left(\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2},n-1}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2},n-1}^2}\right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

Then $(1-\alpha)100\%$ confidence interval for variance when the population mean is unknown is,

$$\left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right]$$

$(1-\alpha)100\%$ Confidence Interval for variance

Ex:-

The following are the weights, in decagrams, of 10 packages of grass seed distributed by a certain company:

46.4, 46.1, 45.8, 47, 46.1, 45.9, 45.8, 46.9, 45.2, 46

Find the 95% confidence interval for the variance of all such packages of grass seed distributed by this company, assuming a normal population.

$$S^2 = 0.2862$$

$$\chi_{0.025,9}^2 = 19.0$$

$$\chi_{0.975,9}^2 = 2.70$$

$$\left[\frac{9 \times 0.2862}{19}, \frac{9 \times 0.2862}{2.70} \right] = [0.135, 0.954]$$

Chi Squared Table

Degrees of Freedom	Area to the Right of Critical Value							
	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01
1	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892

$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

We know that for two samples of two populations where means are unknown,

$$\frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F(n_1 - 1, n_2 - 1)$$

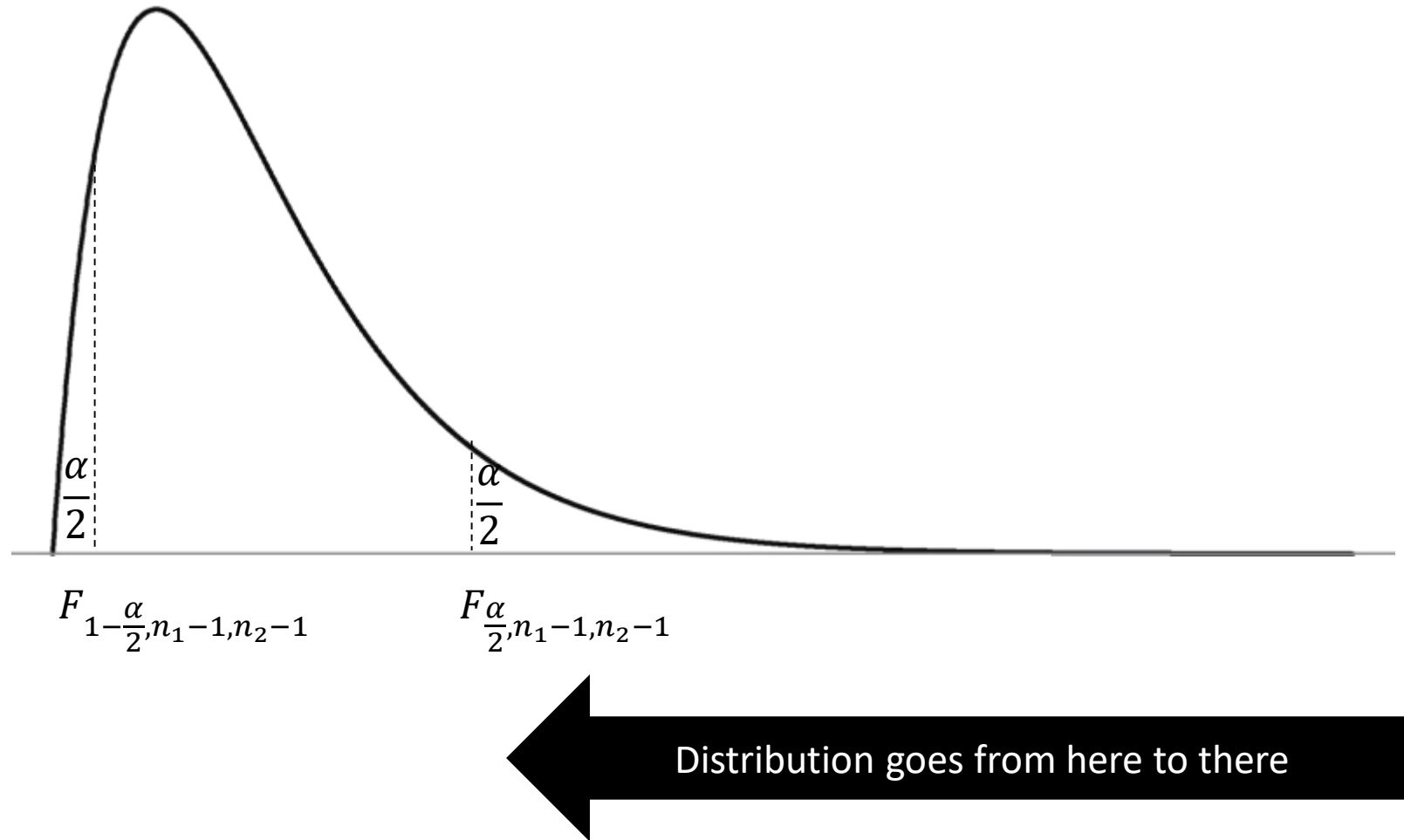
Where n_1 and n_2 are sample sizes and σ_1^2, S_1^2 are variances for the first population and its sample as well as σ_2^2, S_2^2 are variances for the second population and its sample.

Consider this as the pivotal quantity. Then,

$$P\left(L \leq \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \leq U\right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

In the F distribution,



$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

Then,

$$P\left(F_{1-\frac{\alpha}{2}, n_1-1, n_2-1} \leq \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \leq F_{\frac{\alpha}{2}, n_1-1, n_2-1}\right) = 1 - \alpha$$

$$P\left(\frac{S_2^2}{S_1^2} F_{1-\frac{\alpha}{2}, n_1-1, n_2-1} \leq \frac{\sigma_2^2}{\sigma_1^2} \leq \frac{S_2^2}{S_1^2} F_{\frac{\alpha}{2}, n_1-1, n_2-1}\right) = 1 - \alpha$$

Then $(1-\alpha)100\%$ confidence interval for ration of two population variances $\left(\frac{\sigma_2^2}{\sigma_1^2}\right)$ is,

$$\left[\frac{S_2^2}{S_1^2} F_{1-\frac{\alpha}{2}, n_1-1, n_2-1}, \quad \frac{S_2^2}{S_1^2} F_{\frac{\alpha}{2}, n_1-1, n_2-1} \right]$$

$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

Ex:-

Two independent random samples of water bottles were drawn from 2 automatic bottle filling machines in a factory. Volume of water in each selected bottle was measured. 25 bottles from machine 1 had sample variance $S_1^2 = 1.63$ and 16 observations from machine 2 had sample variance $S_2^2 = 1.23$. Construct 90% confidence interval for the ratio $\frac{\sigma_2^2}{\sigma_1^2}$.

$$S_1^2 = 1.63 \quad S_2^2 = 1.23$$

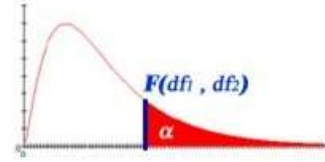
$$n_1 = 25 \quad n_2 = 16$$

$$F_{95\%, 24, 15} = 0.47 \quad F_{5\%, 24, 15} = 2.29$$

$$\frac{S_2^2}{S_1^2} = 0.75$$

$$[0.75 \times 0.47, 0.75 \times 2.29] = [0.3525, 1.7175]$$

F Table



F Table for $\alpha = 0.05$

/	df ₁ =1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
df ₂ =2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

$$F_{\beta, n, m} = \frac{1}{F_{1-\beta, m, n}}$$

Testing one population variance

Consider following hypotheses,

$$H_0 : \sigma^2 = k , H_1 : \sigma^2 \neq k$$

$$H_0 : \sigma^2 = k , H_1 : \sigma^2 > k$$

$$H_0 : \sigma^2 = k , H_1 : \sigma^2 < k$$

Where σ^2 represents the population variance and $k > 0$. Then the test statistic is,

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Testing one population variance

Ex:-

A manufacturer of hard safety hats for construction workers is concerned about the mean and the variation of the forces its helmets transmits to wearers when subjected to an external force. The manufacturer has designed the helmets so that the mean force transmitted by the helmets to the workers is μ pounds (or less) with a standard deviation to be less than 40 pounds. Tests were run on a random sample of 40 helmets and the sample mean and sample standard deviation were found to be 825 pounds and 48.5 pounds respectively. Do the data provide sufficient evidence at 5% level of significance to conclude that the population standard deviation exceeds 40 pounds.

$$H_0 : \sigma^2 = 40^2 = 1600, H_1 : \sigma^2 > 40^2 = 1600$$

Under H_0 ,

$$\chi_{cal}^2 = \frac{39 \times 48.5^2}{1600} = 57.33$$

Testing one population variance

Ex:-

$$\chi_{0.05,39}^2 = 54.57223$$

Since this is a one tail right tail test, since $\chi_{cal}^2 > \chi_{0.05,39}^2$, H_0 will be rejected at 5% level of significance.

Conclusion – There is an evidence to say that the population standard deviation exceeds 40 pounds at 5% level of significance.

Comparison of Two variances

Consider following hypotheses,

$$H_0 : \sigma_x^2 = \sigma_y^2, H_1 : \sigma_x^2 \neq \sigma_y^2$$

$$H_0 : \sigma_x^2 = \sigma_y^2, H_1 : \sigma_x^2 > \sigma_y^2$$

$$H_0 : \sigma_x^2 = \sigma_y^2, H_1 : \sigma_x^2 < \sigma_y^2$$

$\sigma_x^2 = \sigma_y^2$ is equivalent for $\frac{\sigma_x^2}{\sigma_y^2} = 1$.

Where σ_x^2, σ_y^2 represent the population variance for the first population and for the second population respectively. Then the test statistic is,

$$F = \frac{S_x^2}{S_y^2}$$

Compare this F statistic value with the F table for degrees of freedoms (n-1) and (m-1) where n and m are the sample sizes. Rejection criteria is similar to the previous sections.

Comparison of Two variances

Ex:-

The variability in the amount of impurities present in a batch of chemicals used in a particular process depends on the length of time that the process is in operation. Suppose a sample of size 25 is drawn from the normal process which is to be compared to a sample of a new process that has been developed to reduce the variability of impurities.

	Sample 01	Sample 02
n	25	25
S^2	1.04	0.51

Consider the following hypothesis,

$$H_0 : \sigma_1^2 = \sigma_2^2, H_1 : \sigma_1^2 > \sigma_2^2$$

Where σ_1^2 , σ_2^2 are variabilities of the old process and the new process respectively.

Comparison of Two variances

Ex:-

$$F_{cal} = \frac{S_1^2}{S_2^2} = \frac{1.04}{0.51} = 2.04$$

$$F_{5\%,24,24} = 1.98$$

Since this is a one tail right tail test as well as since $F_{cal} > F_{5\%,24,24}$, H_0 will be rejected at 5% level of significance.

Conclusion – There is an evidence to say that the variability in the new process is less than the variability in the original process at 5% level of significance.